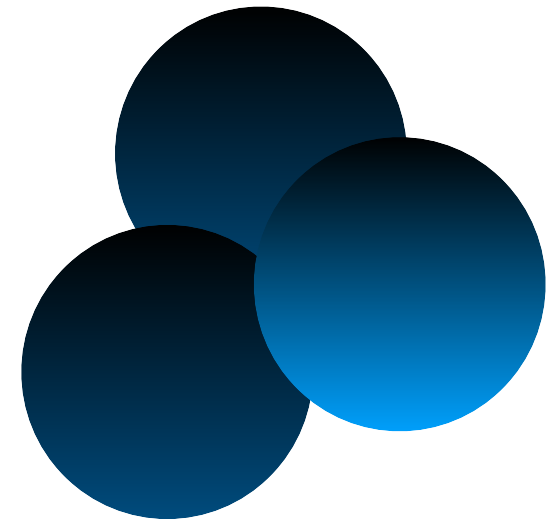




PPC 2024

# Breaking into the window of primordial black hole dark matter with x-ray microlensing

Based on arXiv: 2405.20365 (Submitted to Phys. Rev. Lett. [PRL])



Authors: **Manish Tamta**, Nirmal Raj, Prateek Sharma  
(Ph.D. student at IISc Bangalore)



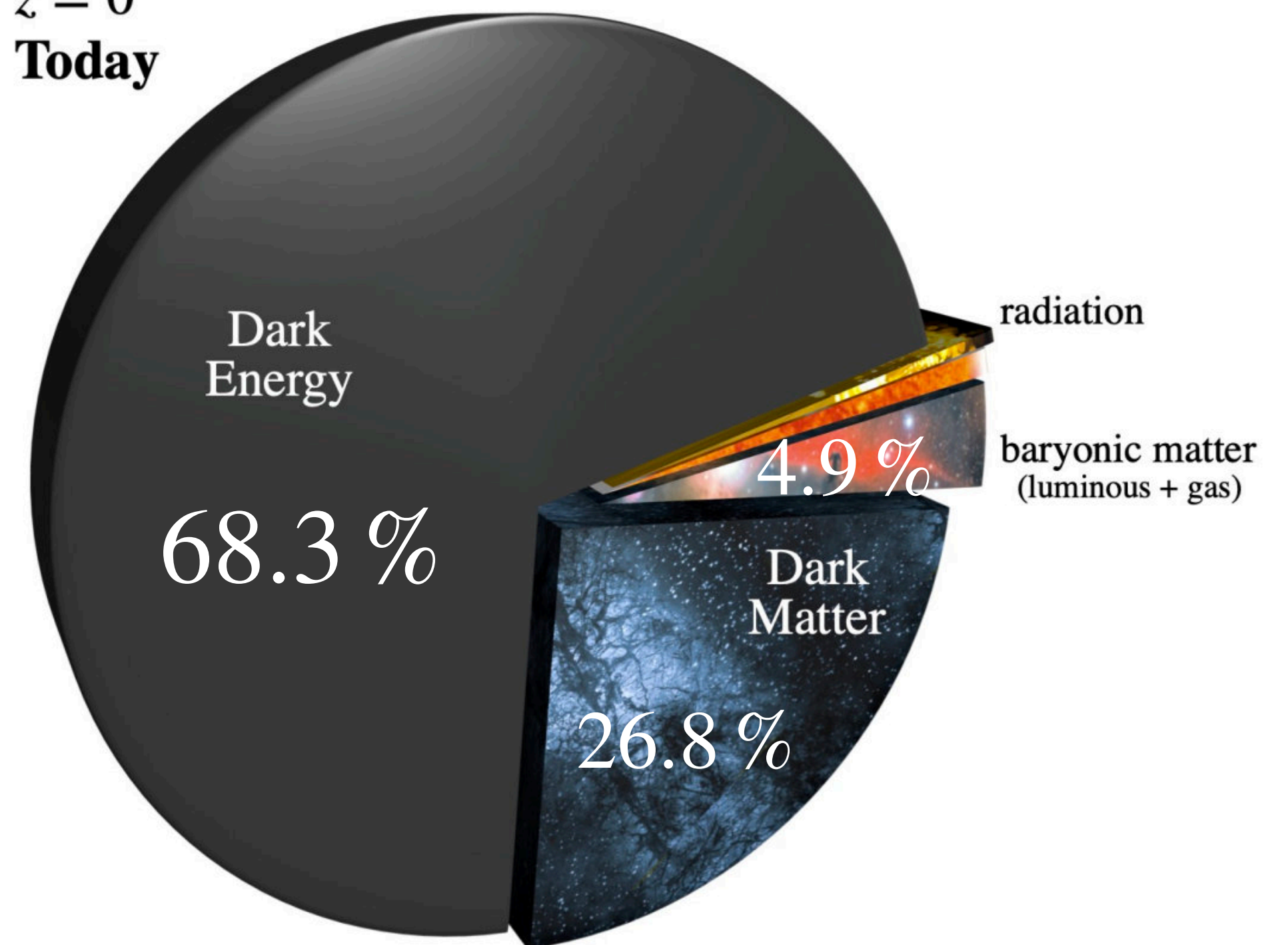
# Ingredients of our universe

What our universe is made up of?

This is what CMB tells  
about the universe we  
live in



$z = 0$   
Today

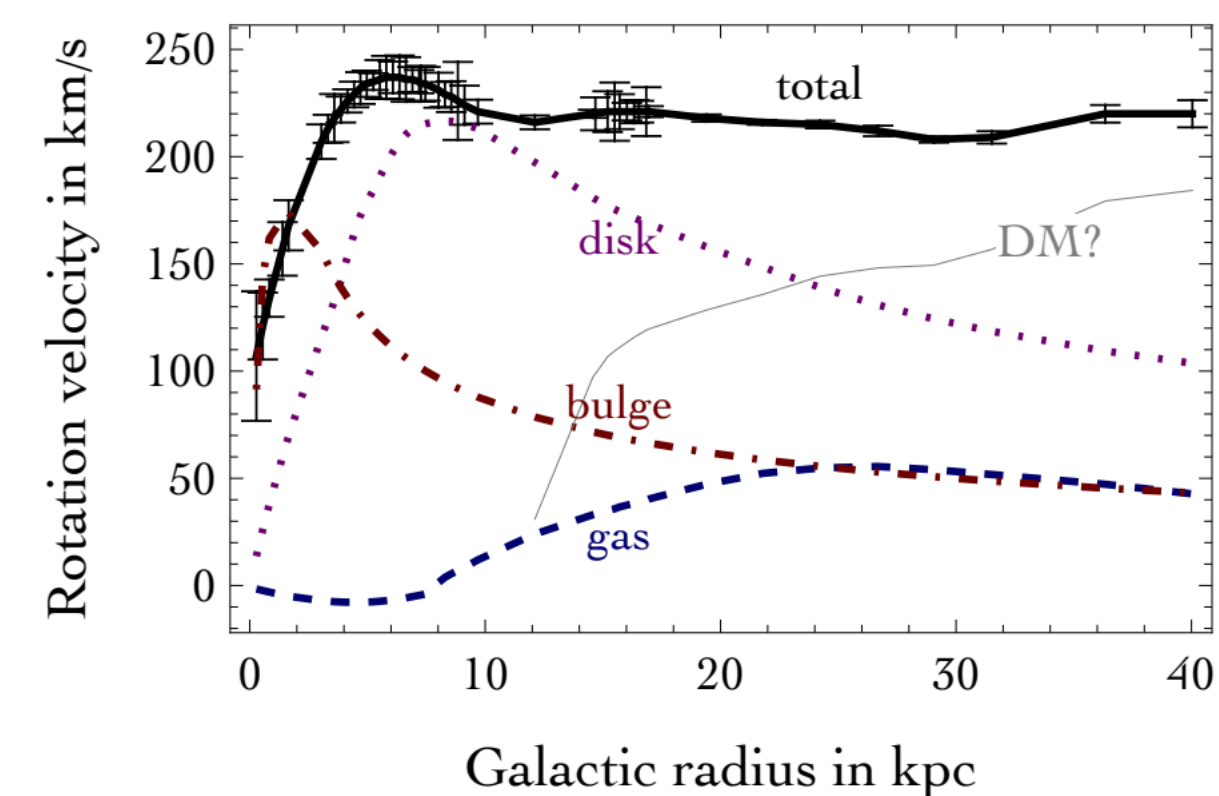


Cirelli et. al. [arXiv: 2406.01705]

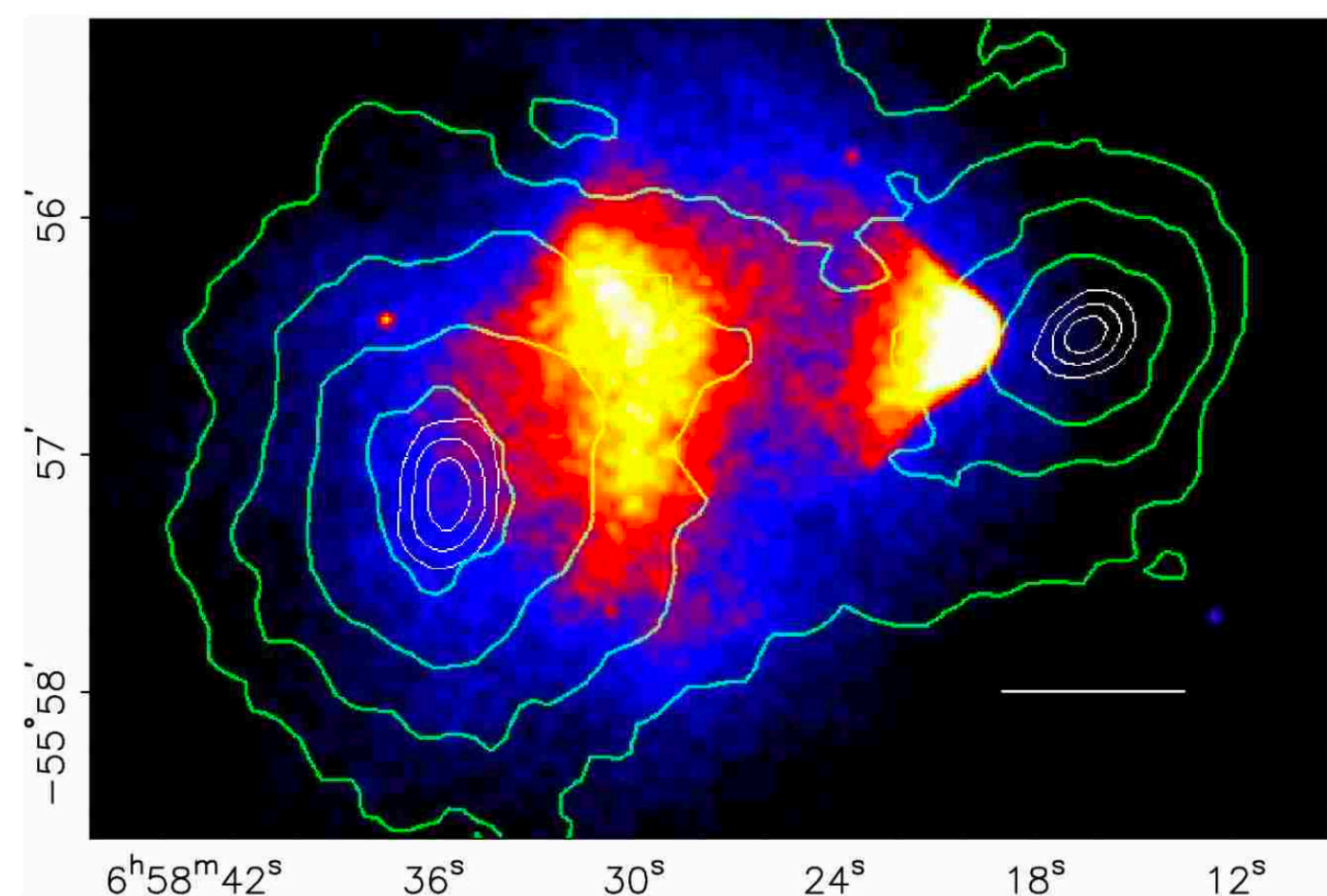
**Dark matter comprises ~85% of total matter in the universe!**



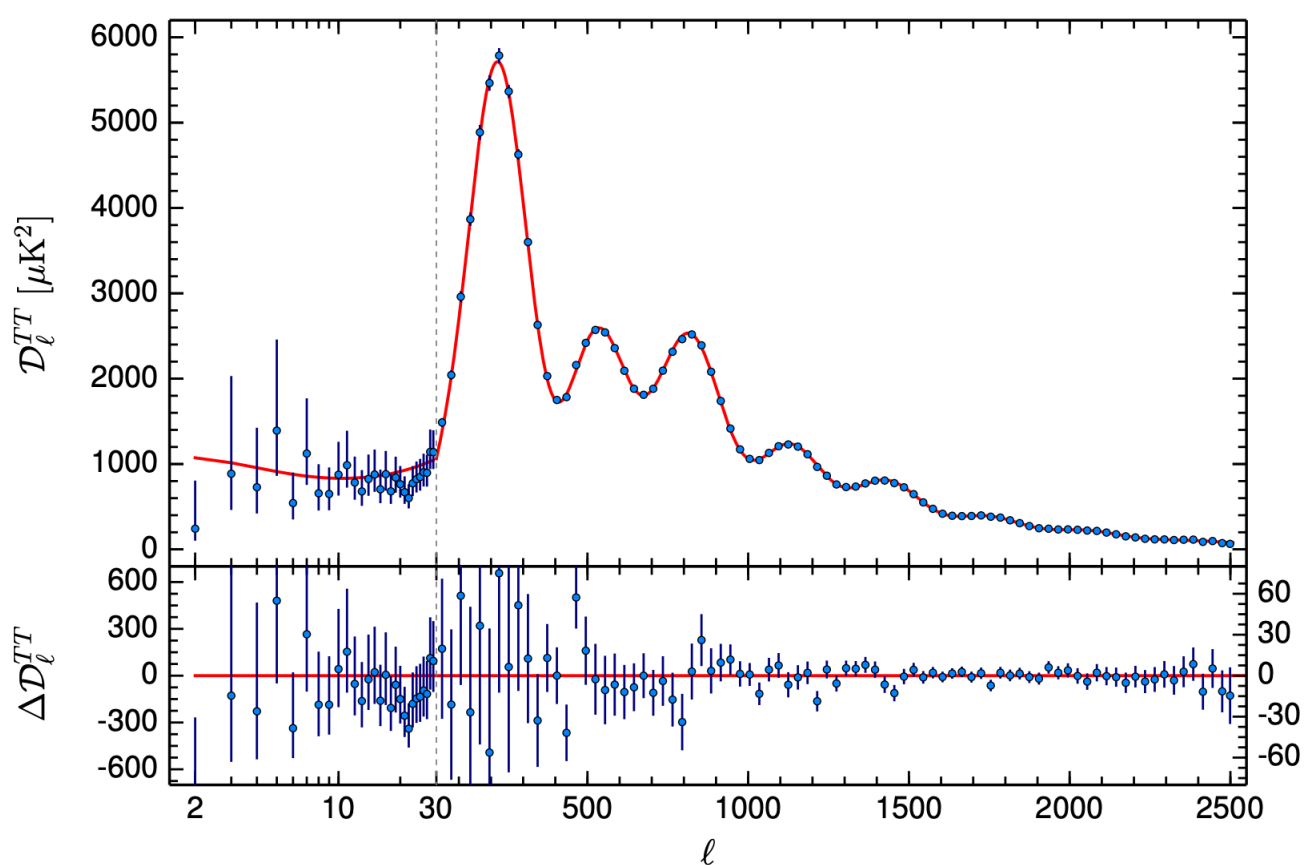
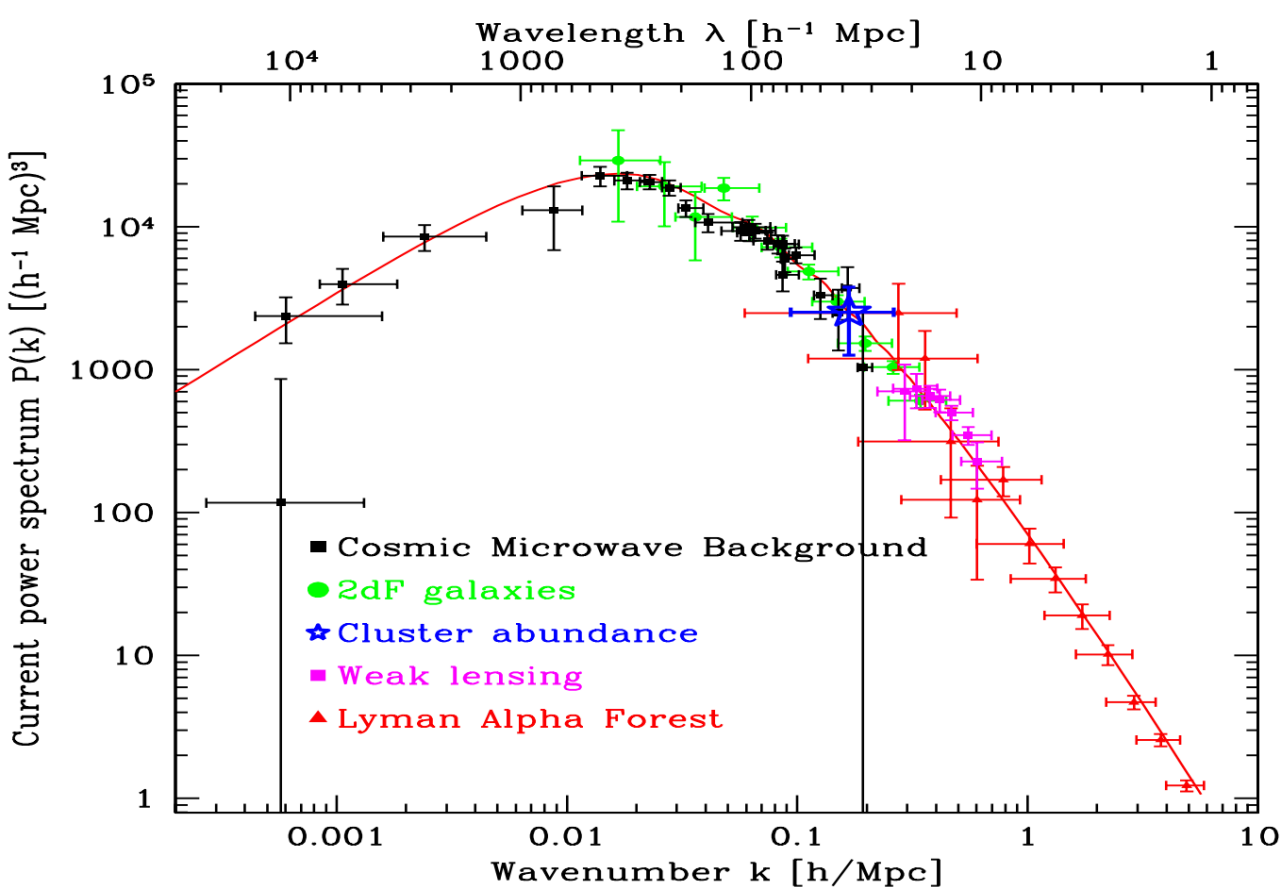
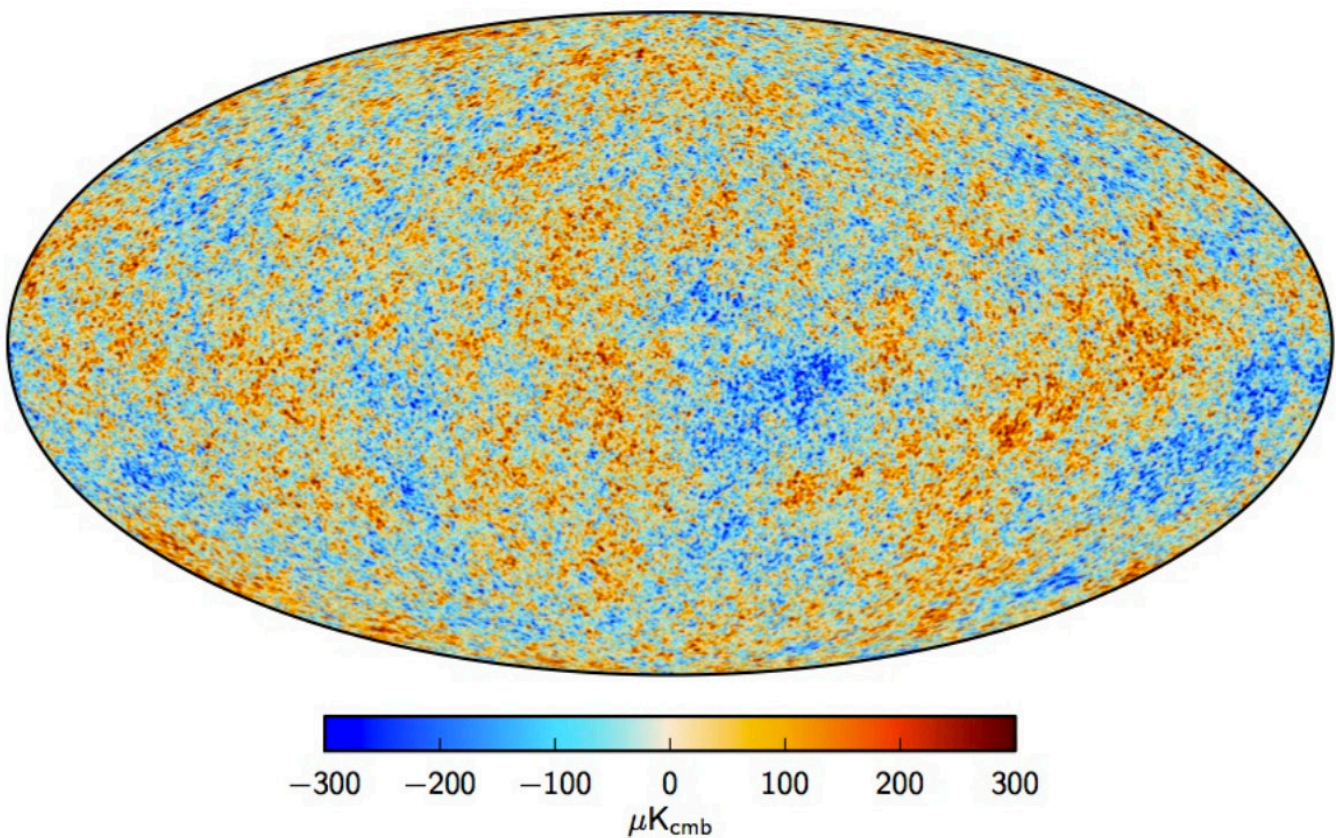
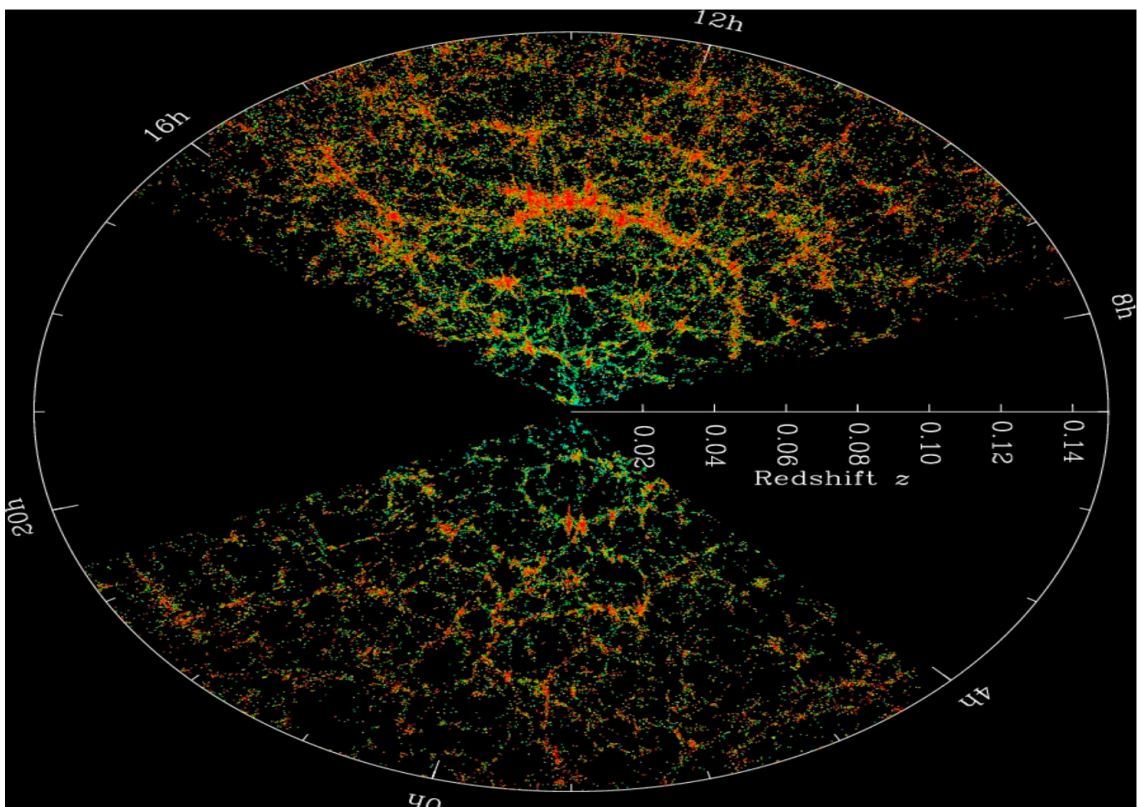
# Evidences of dark matter in the universe



## Smoking gun...Bullet cluster



Clowe et al., The Astrophysical Journal + 2006



Cirelli et. al. [arXiv: 2406.01705]

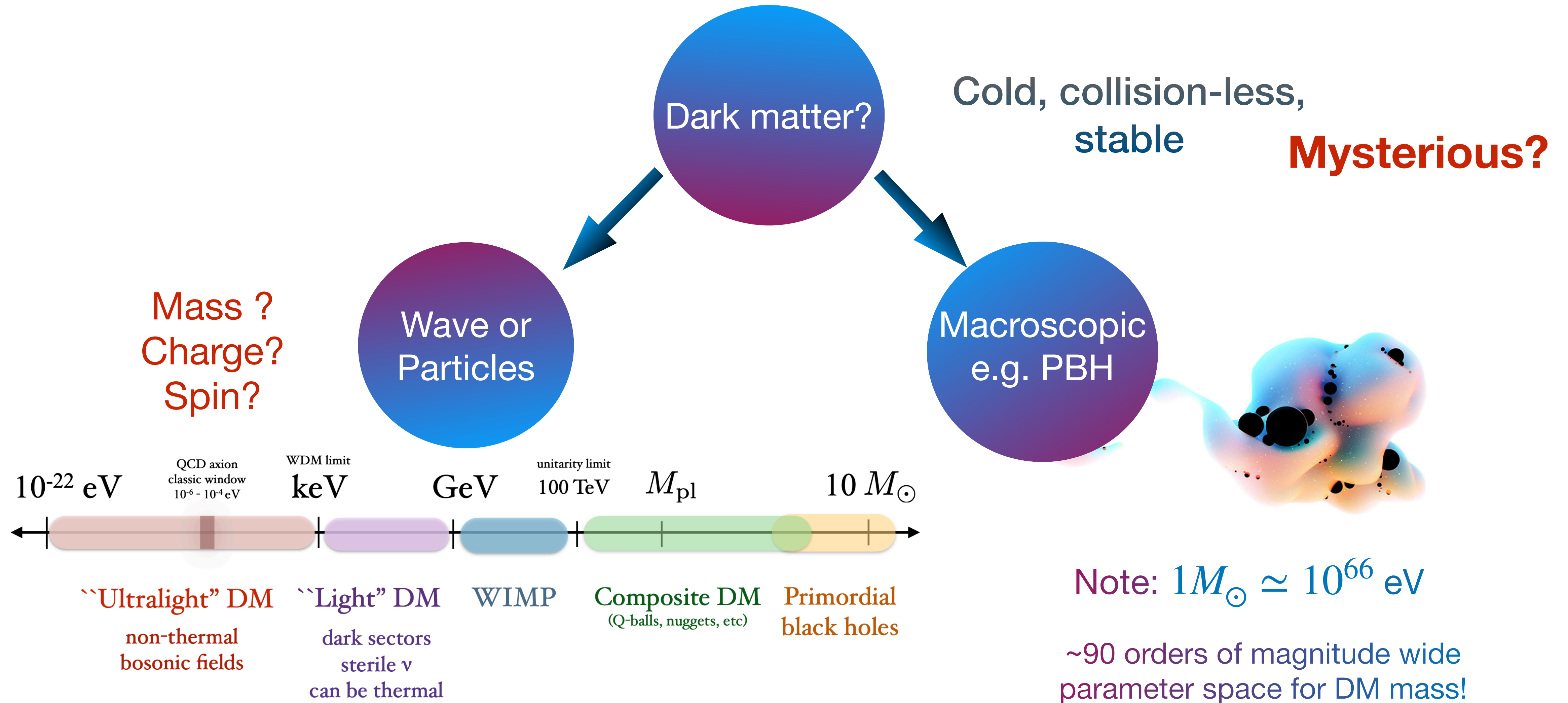
Small scale



Large scale



# What is Dark Matter after all?



[Image: Tongyan Lin + 2019]

Manish Tamta, IISc Bangalore, PPC 2024, [manishtamta@iisc.ac.in](mailto:manishtamta@iisc.ac.in)

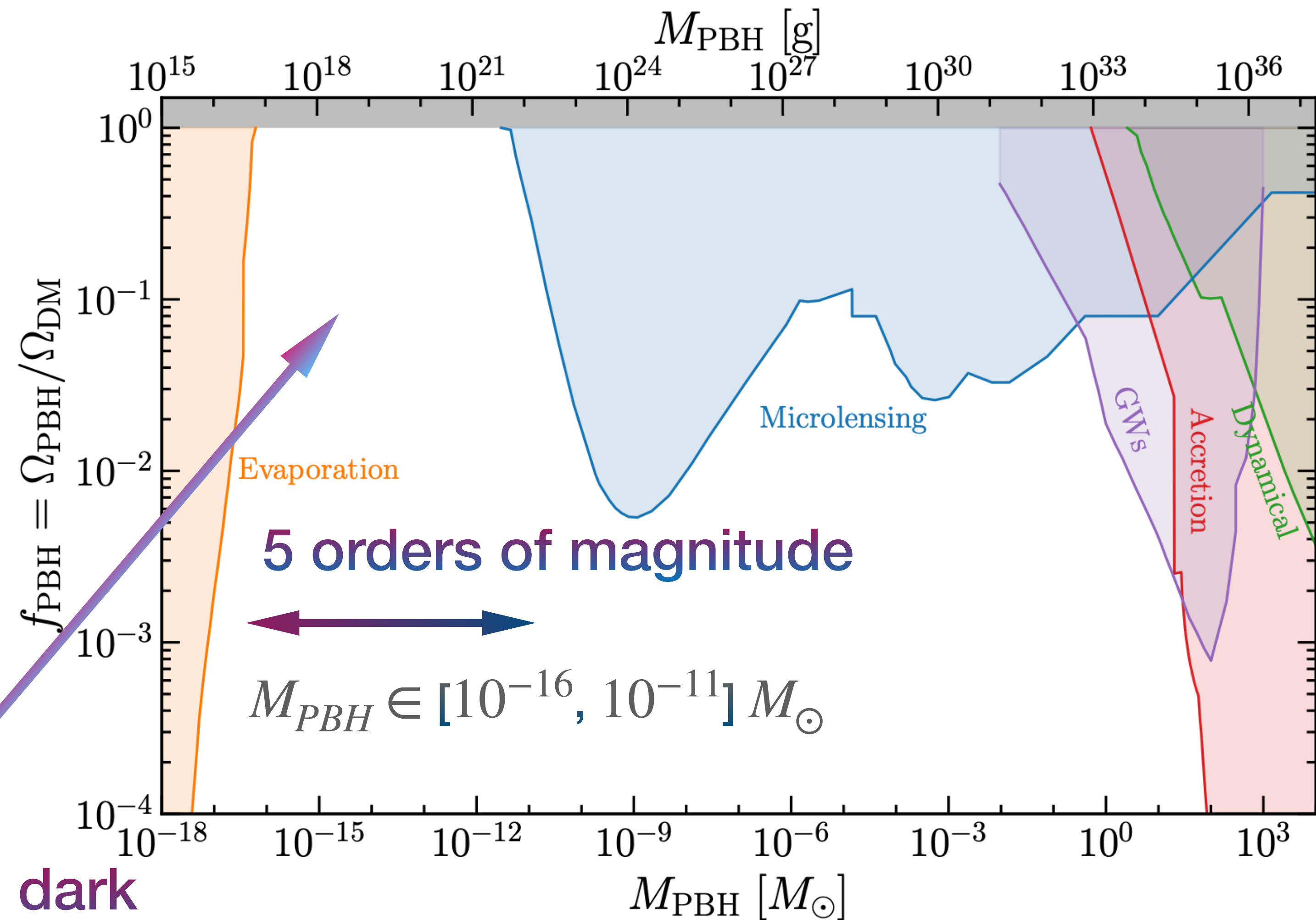
# The primordial black hole dark matter window

PBHs may form due to density perturbations generated during the inflationary epoch!

1974



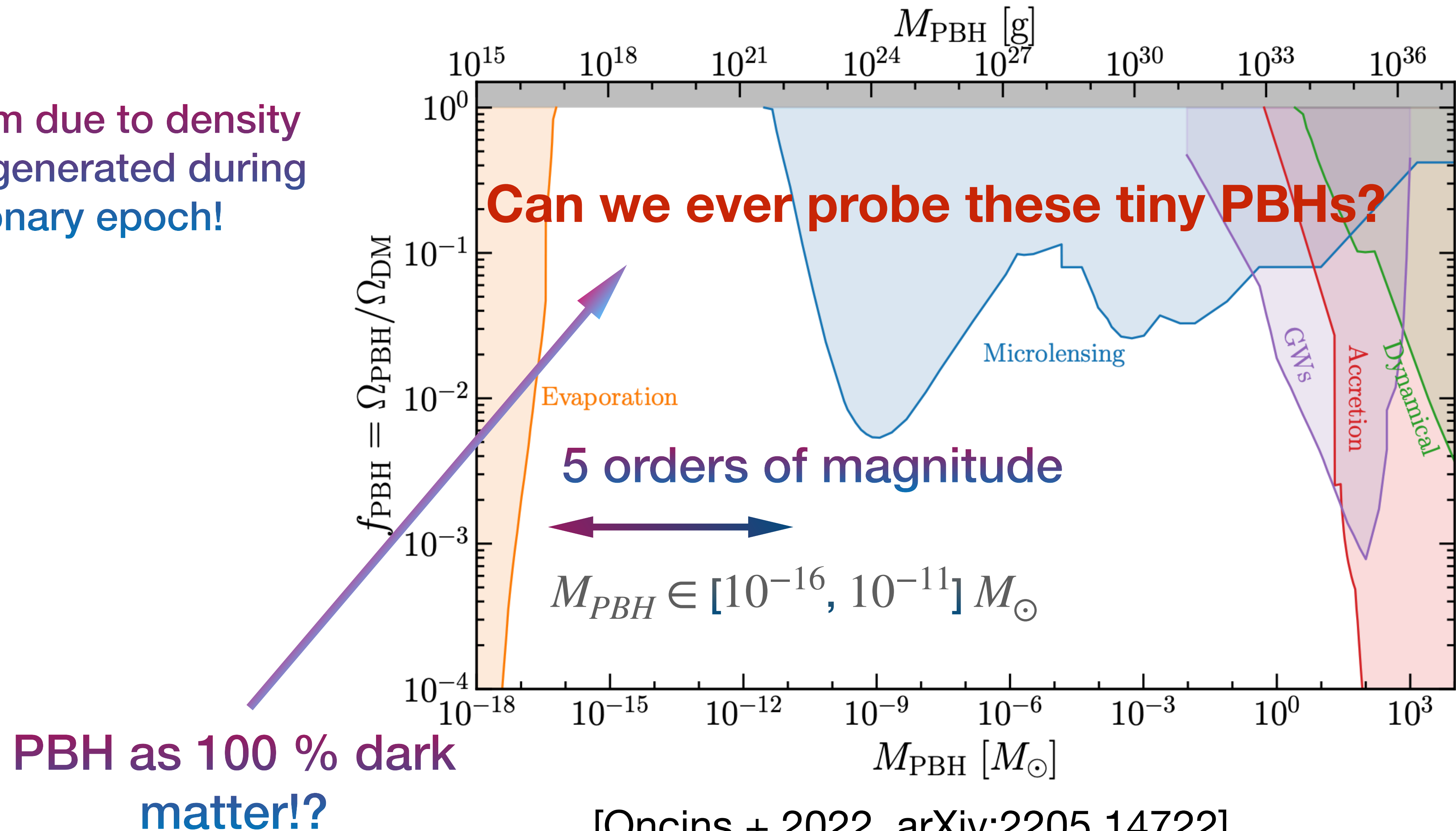
PBH as 100 % dark matter!?



[Oncins + 2022, arXiv:2205.14722]

# The primordial black hole dark matter window

PBHs may form due to density perturbations generated during the inflationary epoch!

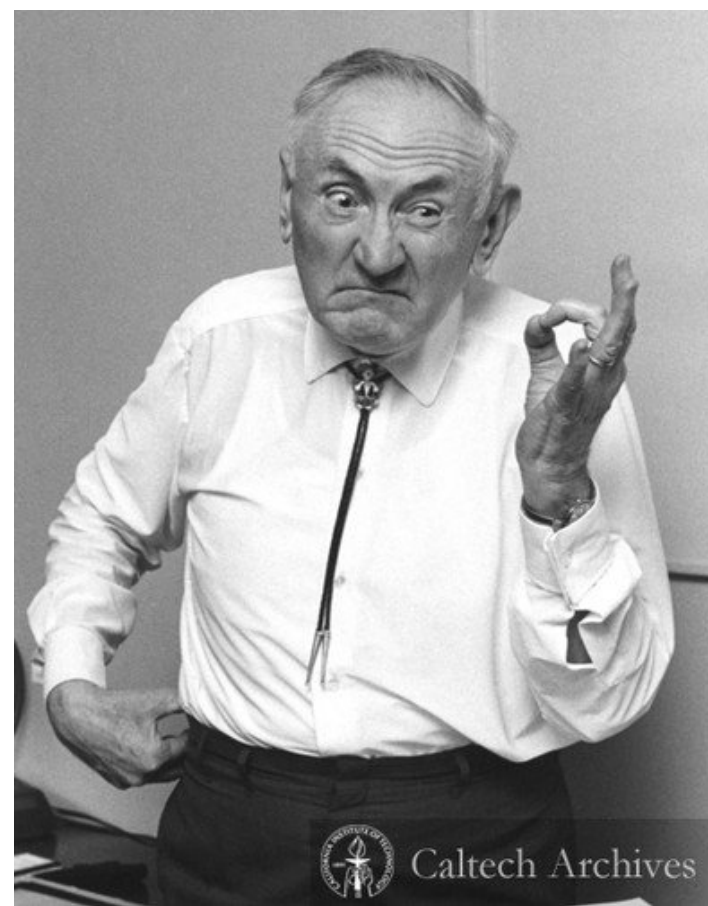


[Oncins + 2022, arXiv:2205.14722]



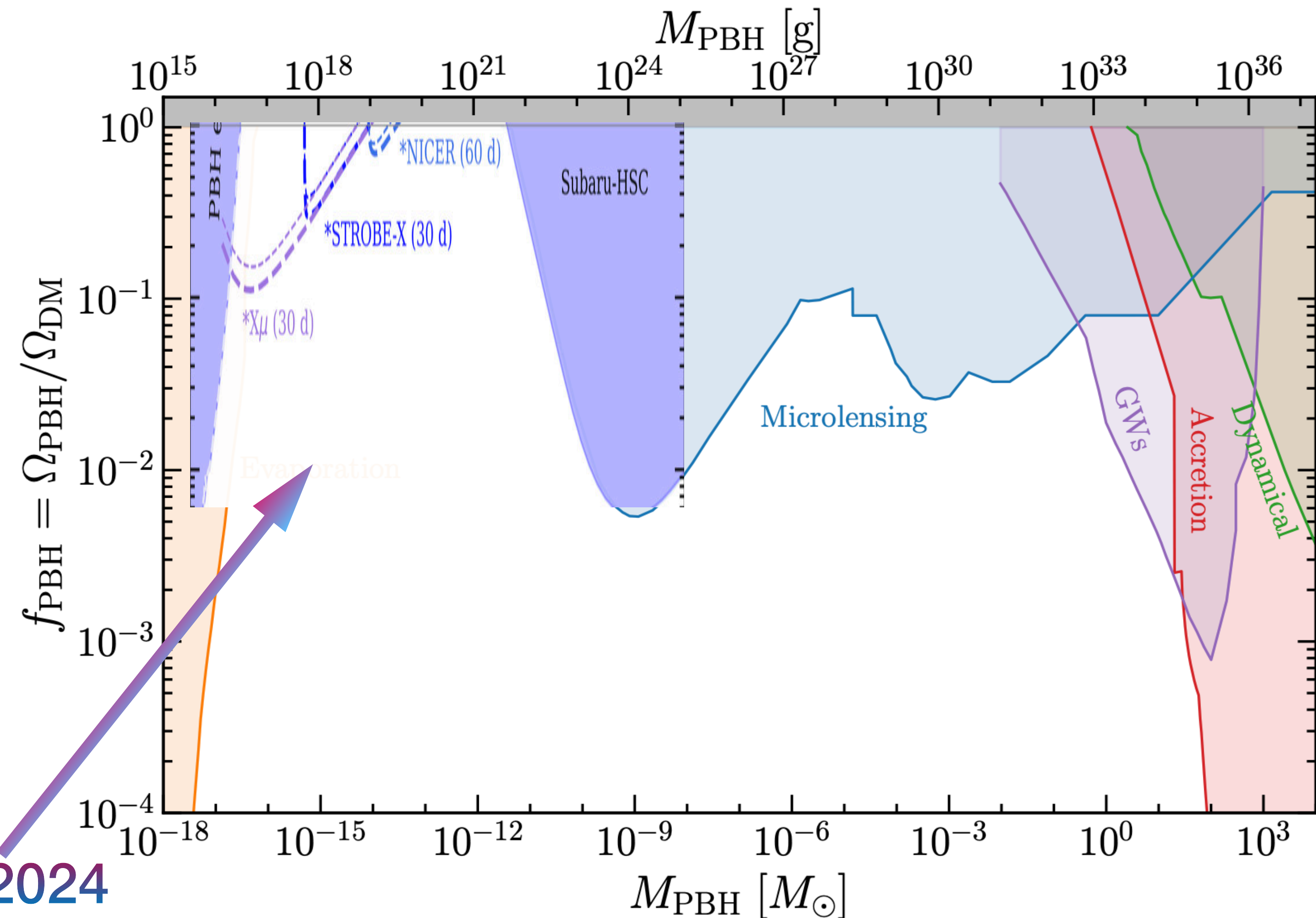
# The primordial black hole dark matter window

Current and future x-ray telescopes can probe this window!

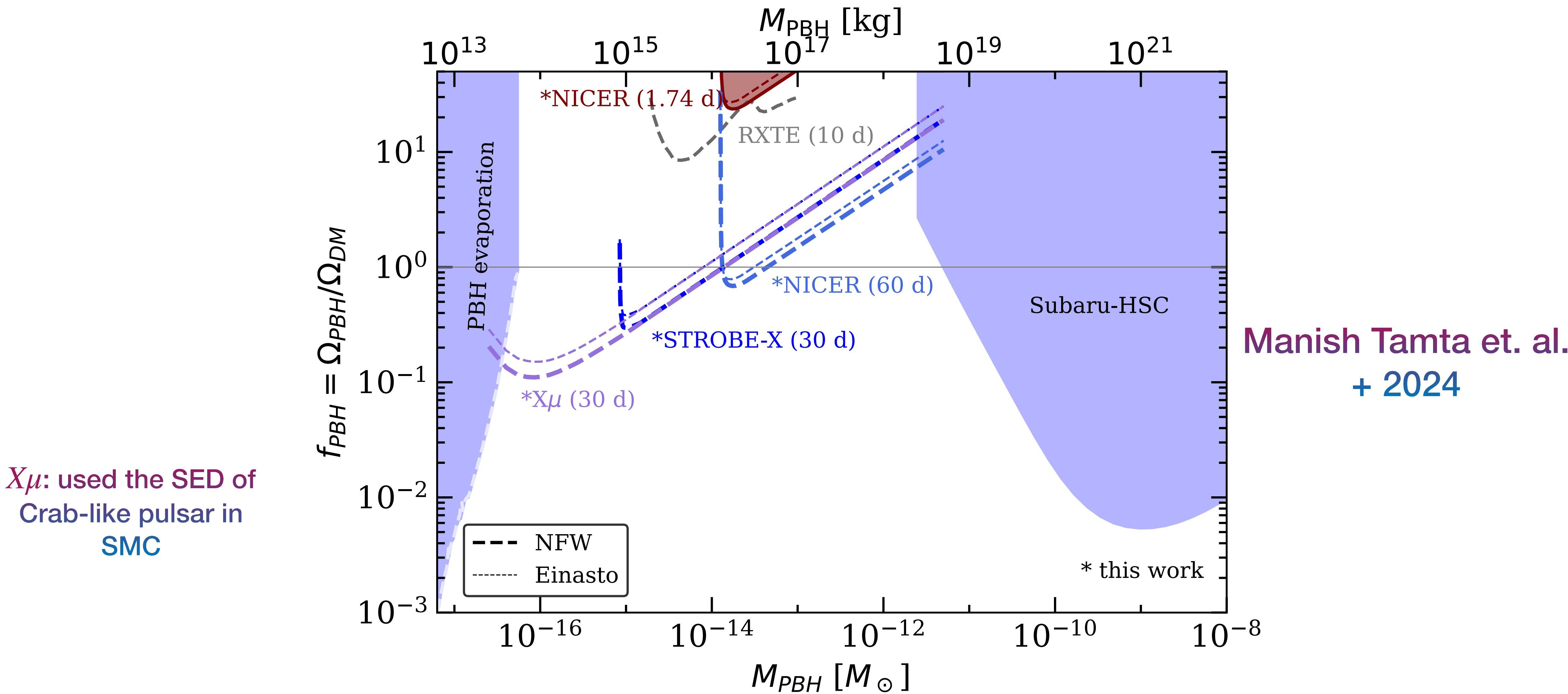


Manish Tamta et. al. + 2024

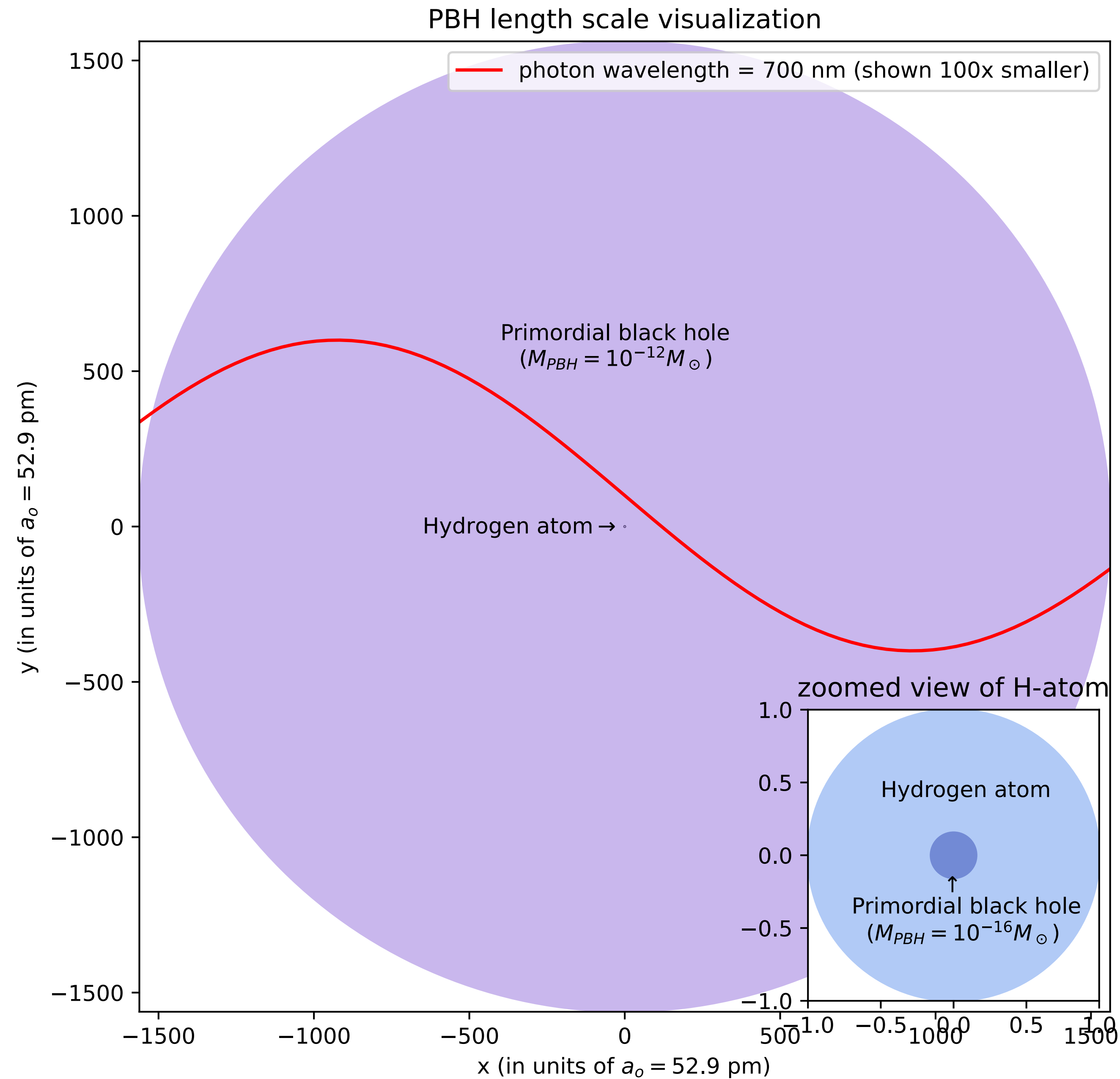
arXiv:2405.20365



# X-ray telescope sensitivities for PBH dark matter detection/exclusion





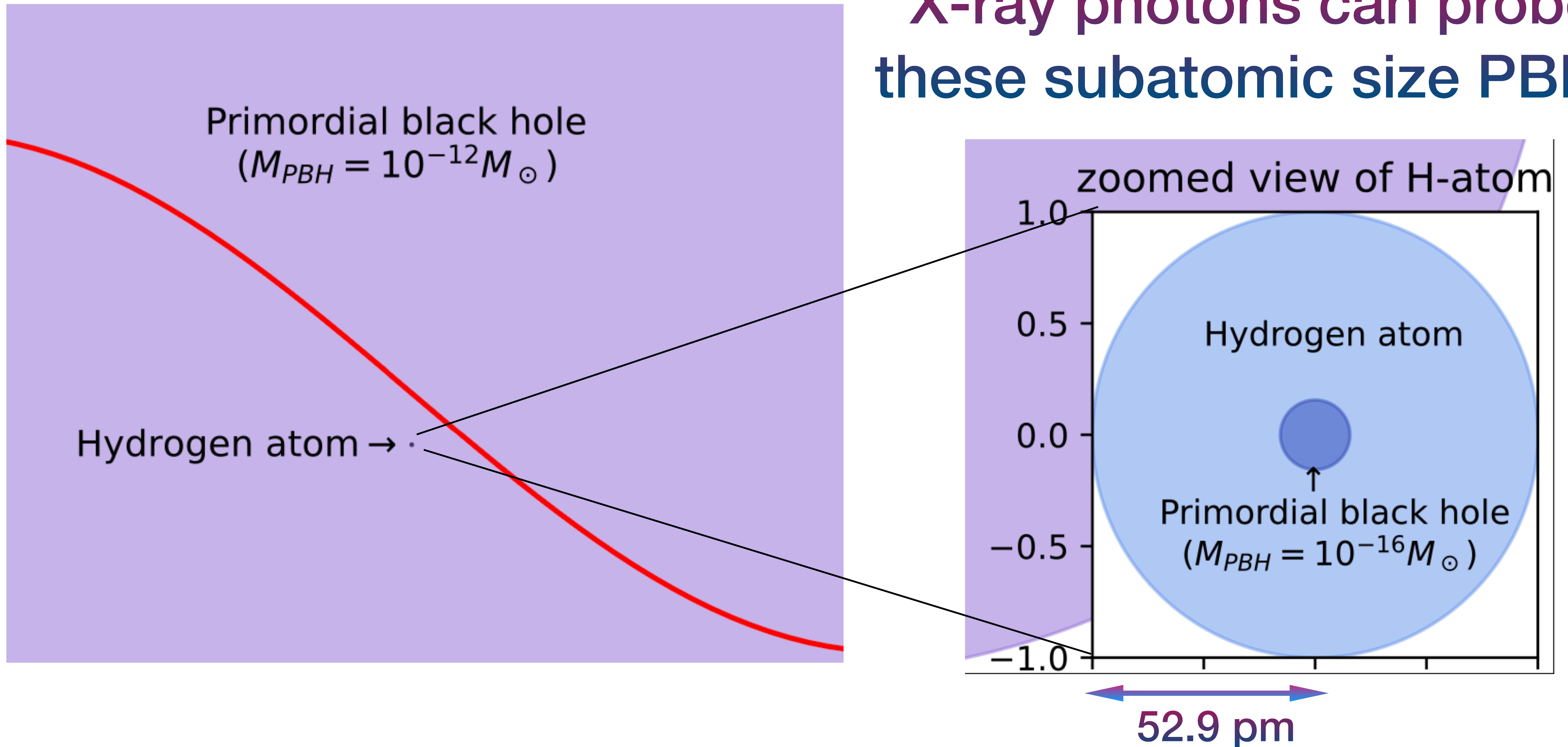


# Mass/length scales involved

$$M_{PBH} \in [10^{-16}, 10^{-11}] M_{\odot}$$

$$R_{Schw} \in [2.9 \times 10^{-13}, 2.9 \times 10^{-8}] m$$

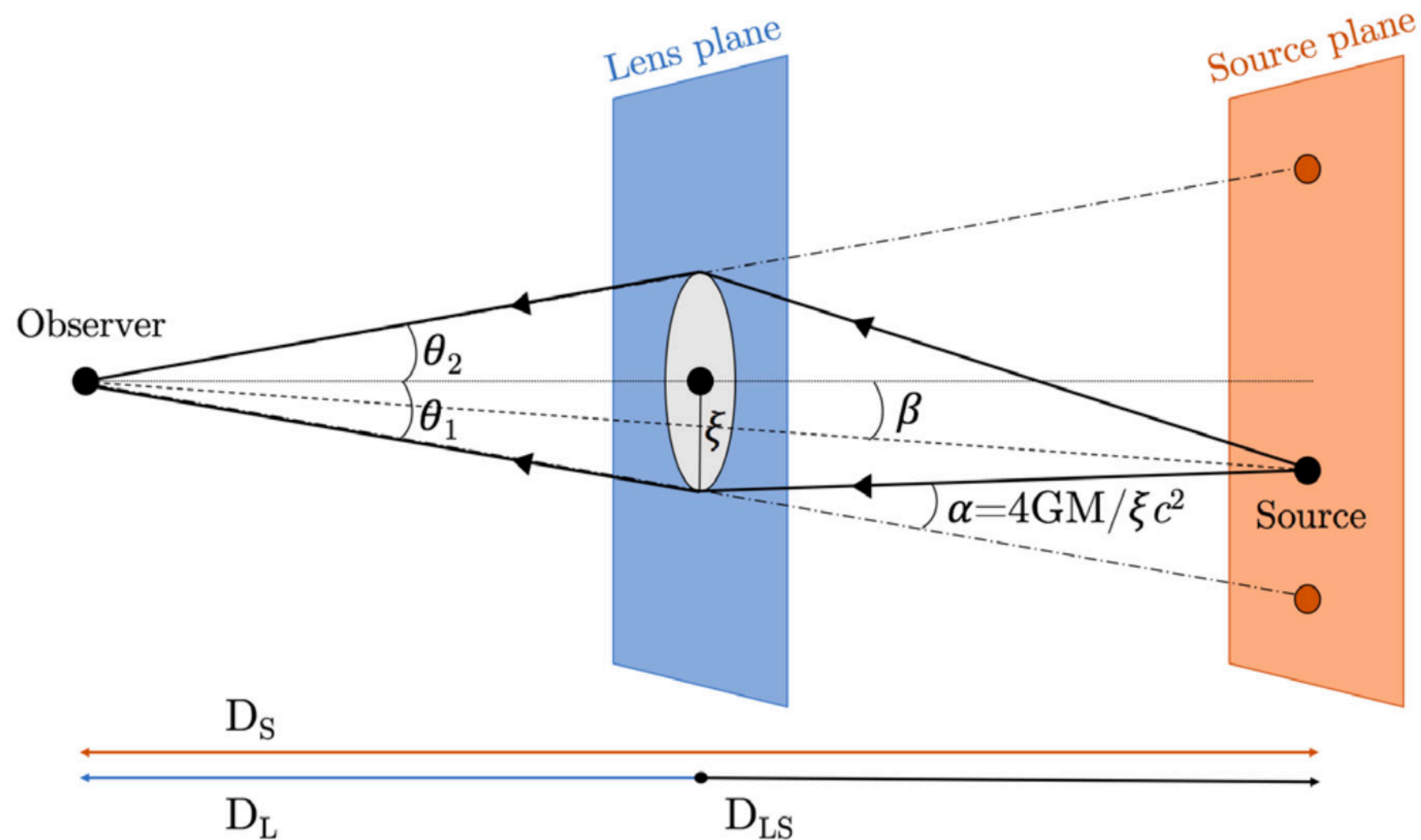
## X-ray photons can probe these subatomic size PBHs



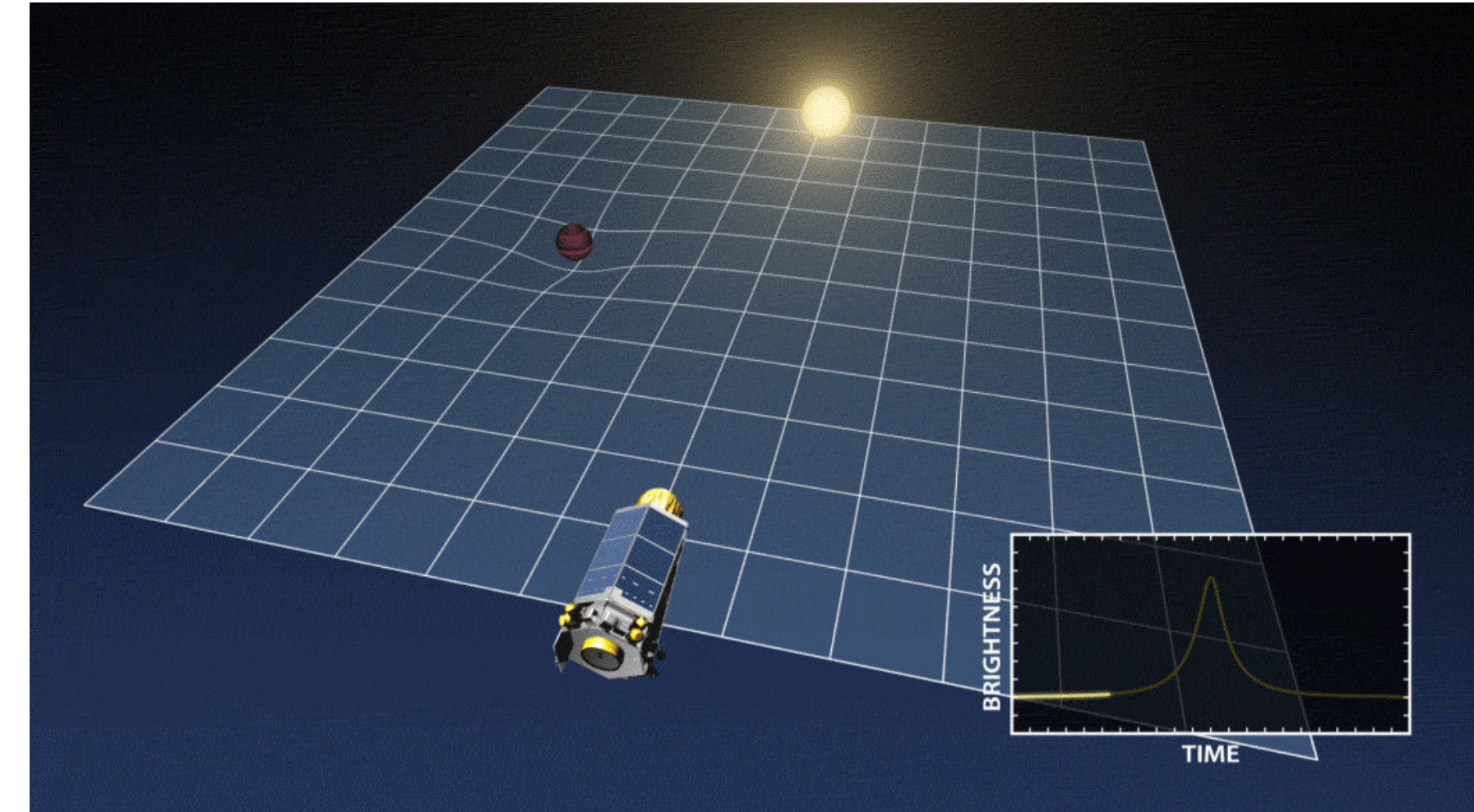


# What is gravitational microlensing

Lens equation:  $\vec{\beta} = \vec{\theta} - \vec{\alpha}$



Croon, McKeen, Raj + 2020



$$\hat{\alpha} = \frac{4GM}{c^2 \xi} = \frac{2R_{Schw}}{\xi}$$

Magnification due to a point lens

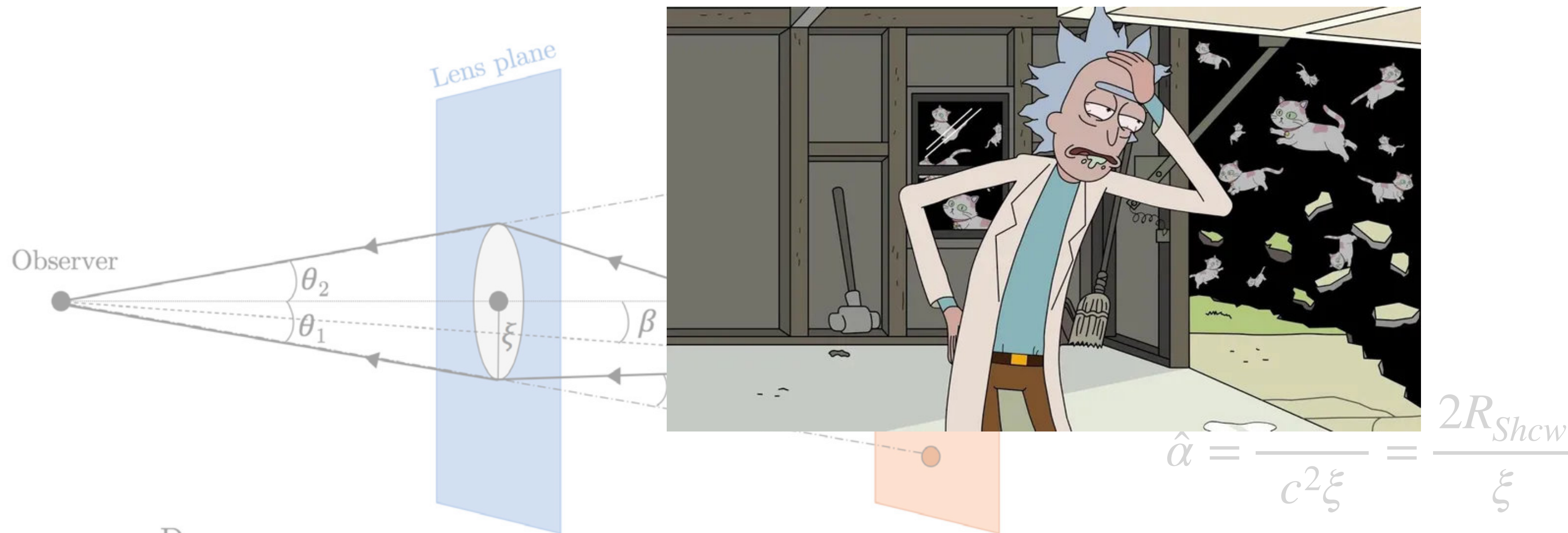
$$\mu = \frac{\theta}{\beta} \frac{d\theta}{d\beta} = \frac{y^2 + 2}{y\sqrt{y^2 + 4}}$$

$$y \equiv \frac{\beta}{\theta_E}$$



# Need for wave optics microlensing

Lens equation:  $\vec{\beta} = \vec{\theta} - \vec{\alpha}$



# Lens equation breaks down when microlensing happens in the wave regime!

Croon, McKeen, Raj + 2020

Manish Tamta, IISc Bangalore, PPC 2024, [manishtamta@iisc.ac.in](mailto:manishtamta@iisc.ac.in)

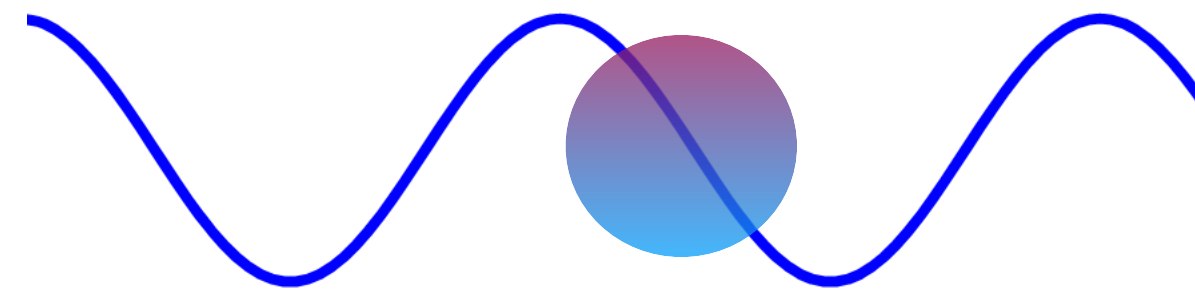


# Wave regime lensing and Parameter $w$

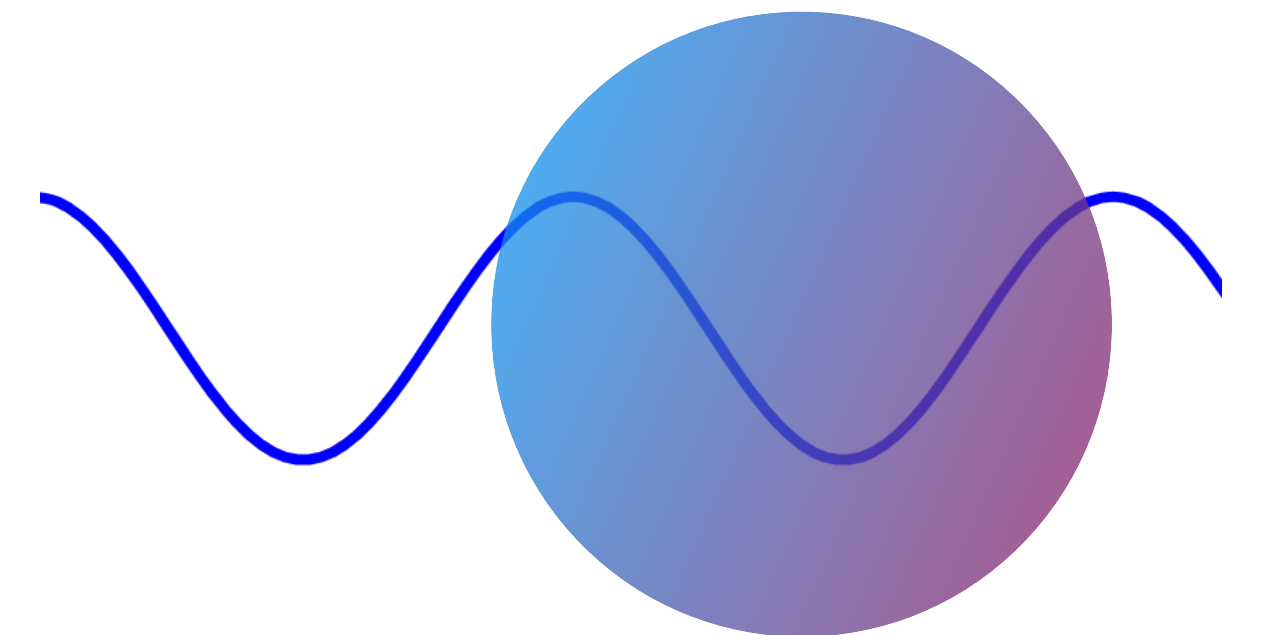
$$w \equiv \frac{4GME_\gamma}{\hbar c^3} = \frac{2R_{schw}E_\gamma}{\hbar c}$$

← Dimensionless frequency

Wave regime: if  $w \lesssim y^{-1}$



Geometric optics regime: when  $w \gg y^{-1}$



**Note that  $y$  is impact parameter!**

# Theory of gravitational microlensing in the wave regime+FSE

$$\bar{\mu}_E(y, a_s(x)) = \frac{1}{\mathcal{N}} \int_{E_{min}}^{E_{max}} dE \mathcal{A}(E) \mathcal{F}(E) a_s^{-2} e^{-\frac{y^2}{2a_s^2}} \frac{\pi w}{1 - e^{-\pi w}} \int_0^\infty r e^{\frac{-r^2}{2a_s^2}} I_0(yr/a_s^2) \left| {}_1F_1\left(\frac{iw}{2}, 1; \frac{iwr^2}{2}\right) \right|^2 dr$$

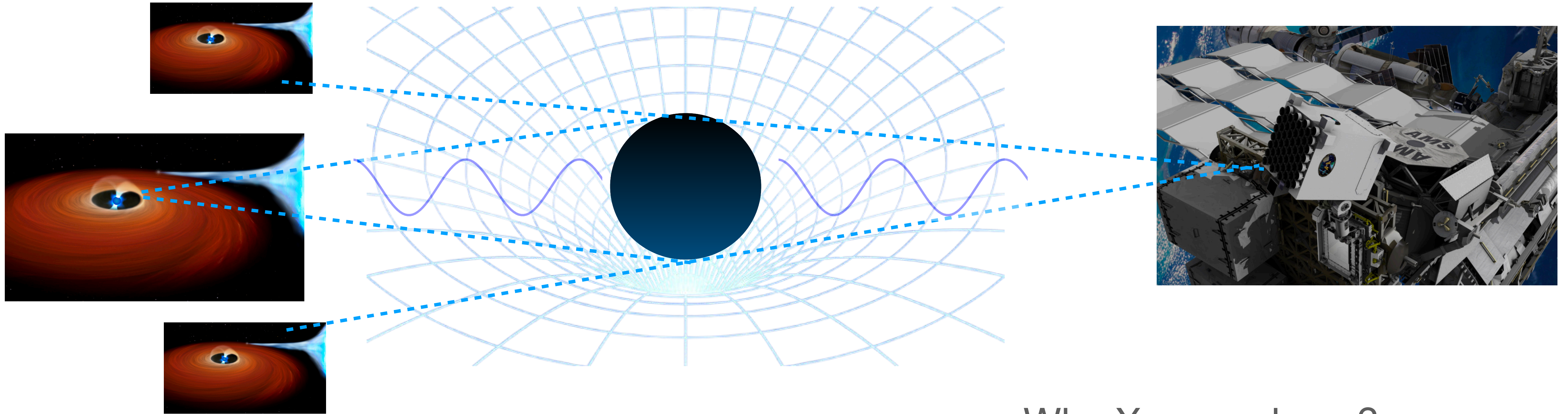
Pulsar  
 Detector  
 Finite extent of the source  
 Wave-effects

$a_s = \frac{xR_S}{r_E}$   
 where  $\mathcal{N} = \int_{E_{min}}^{E_{max}} dE \mathcal{A}(E) \mathcal{F}(E)$

**Frequency dependent magnification!**



# X-ray microlensing can probe subatomic size PBHs



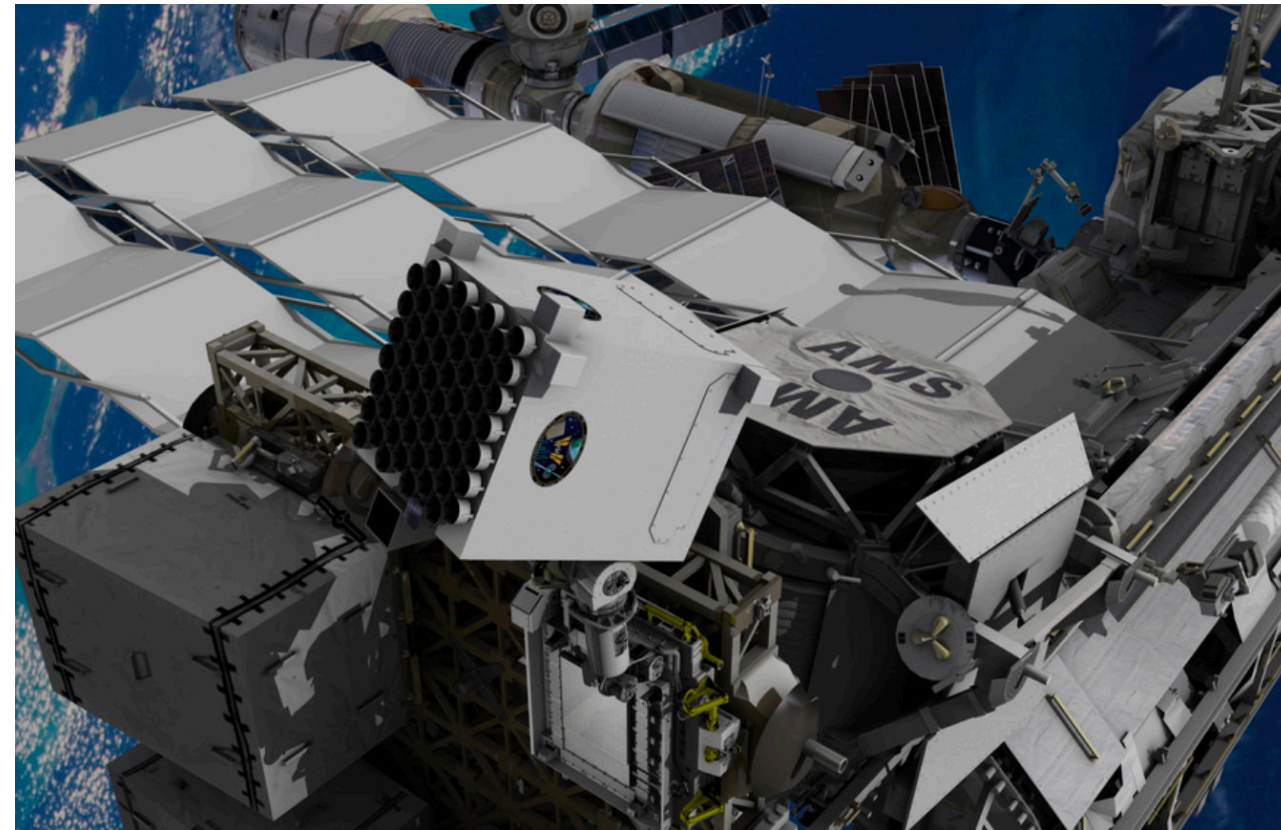
Why X-ray pulsars?

Yang Bai et. al. + 2019

- Less variability and persistent x-ray emission
- Higher photon counts

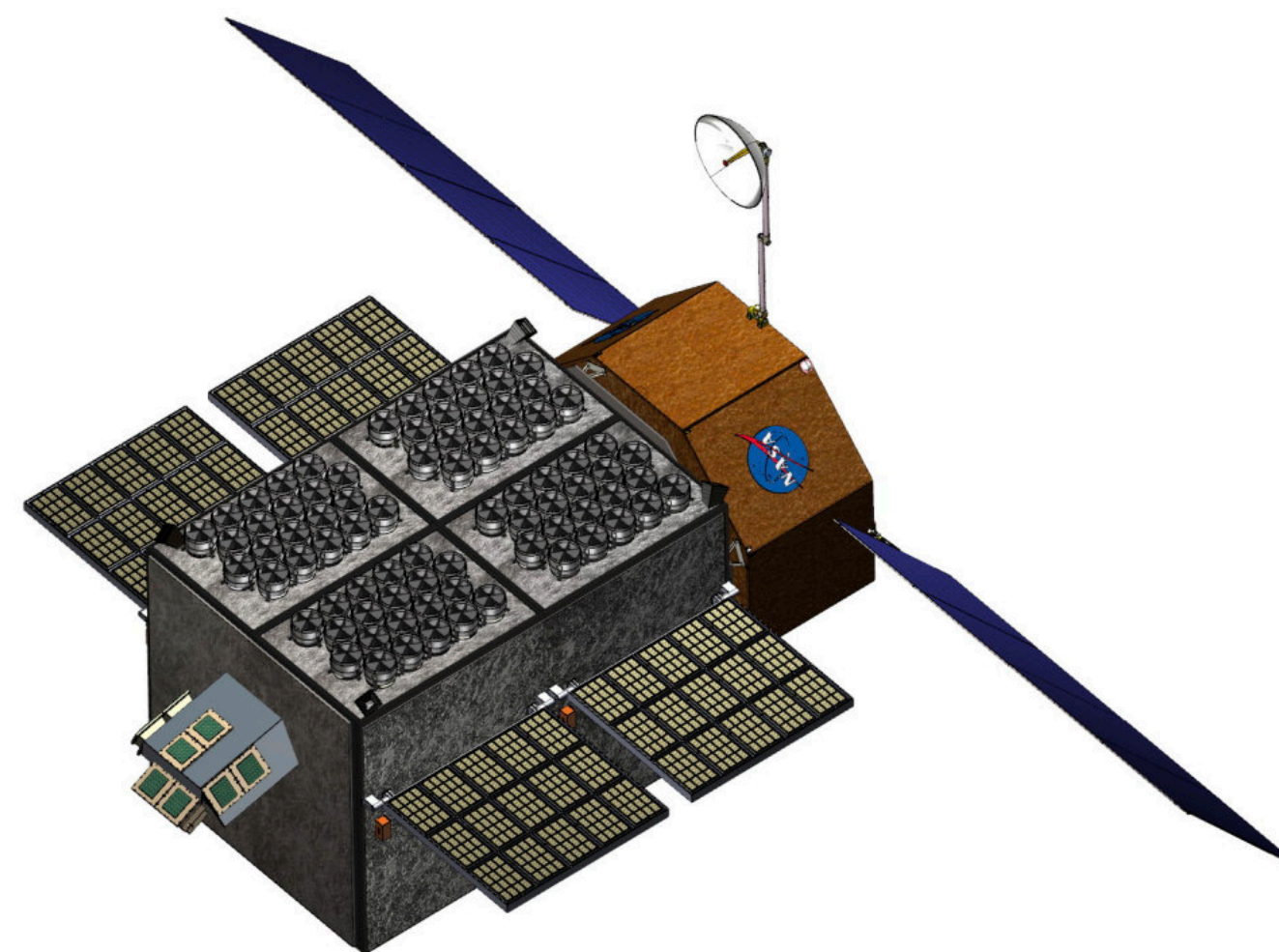


# Telescopes and pulsars for x-ray microlensing



NASA's NICER

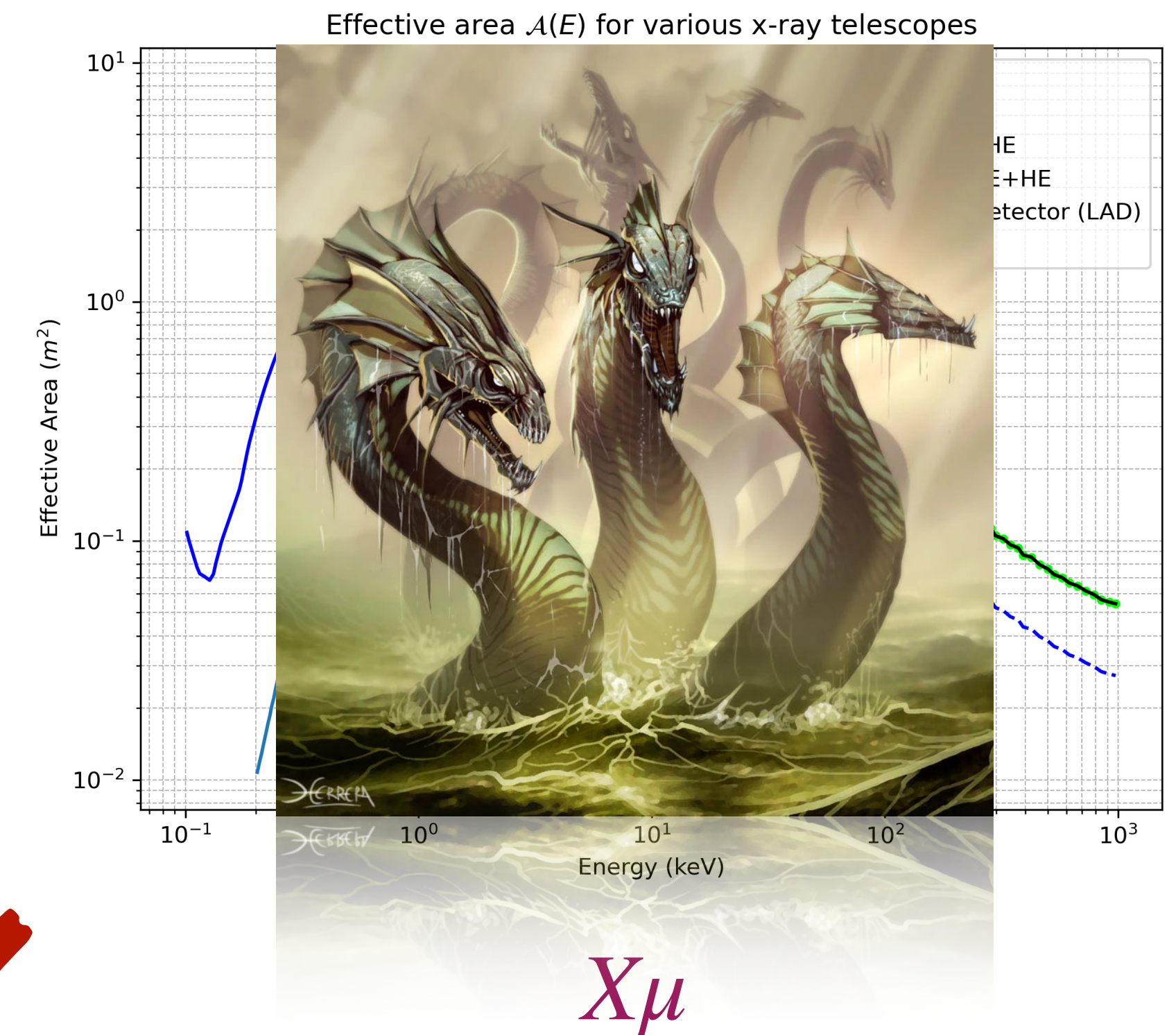
<https://svs.gsfc.nasa.gov/>



STROBE-X

arXiv:1903.03035

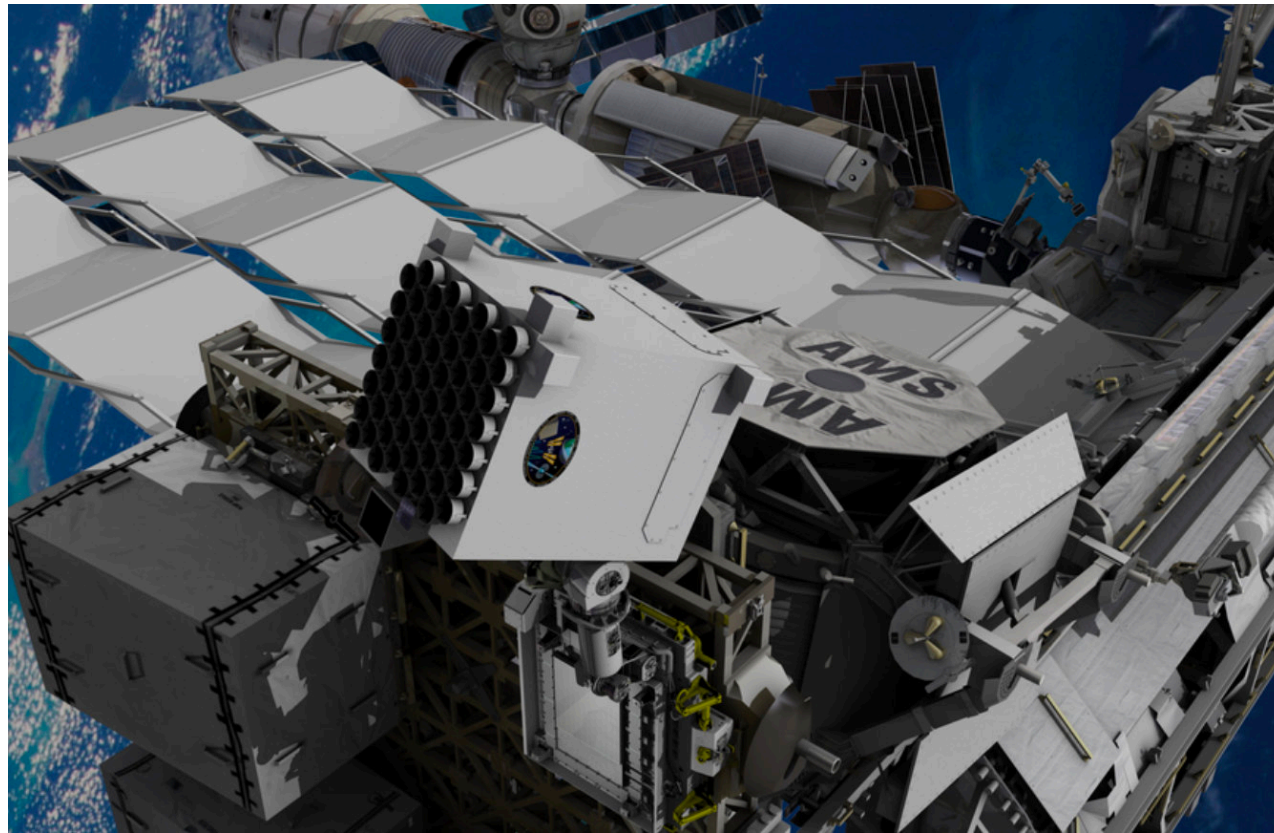
Expected launch: 2031



Let's hope for such  
telescopes in the near future :)

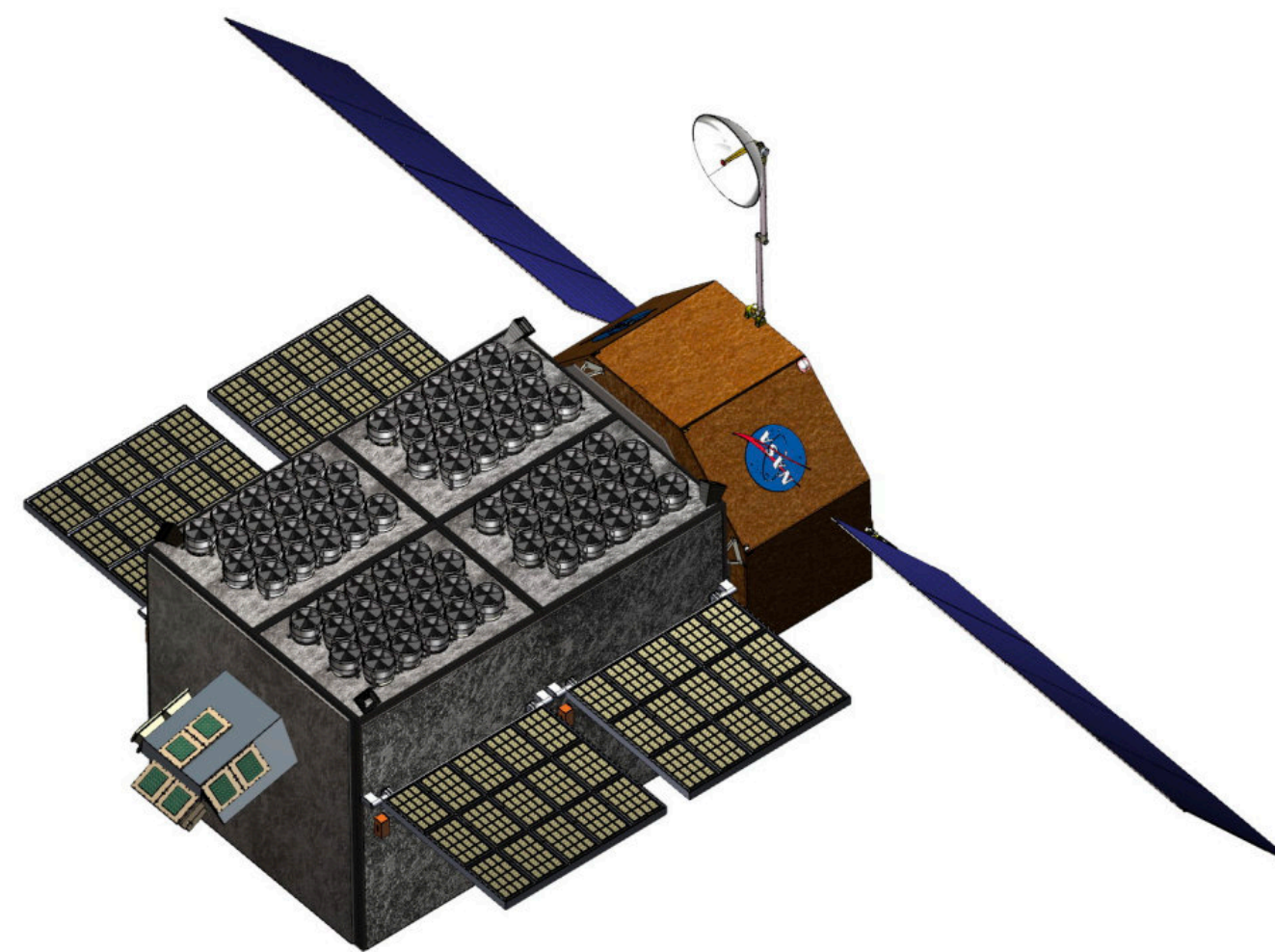


# Telescopes and pulsars for x-ray microlensing



**NASA's NICER**

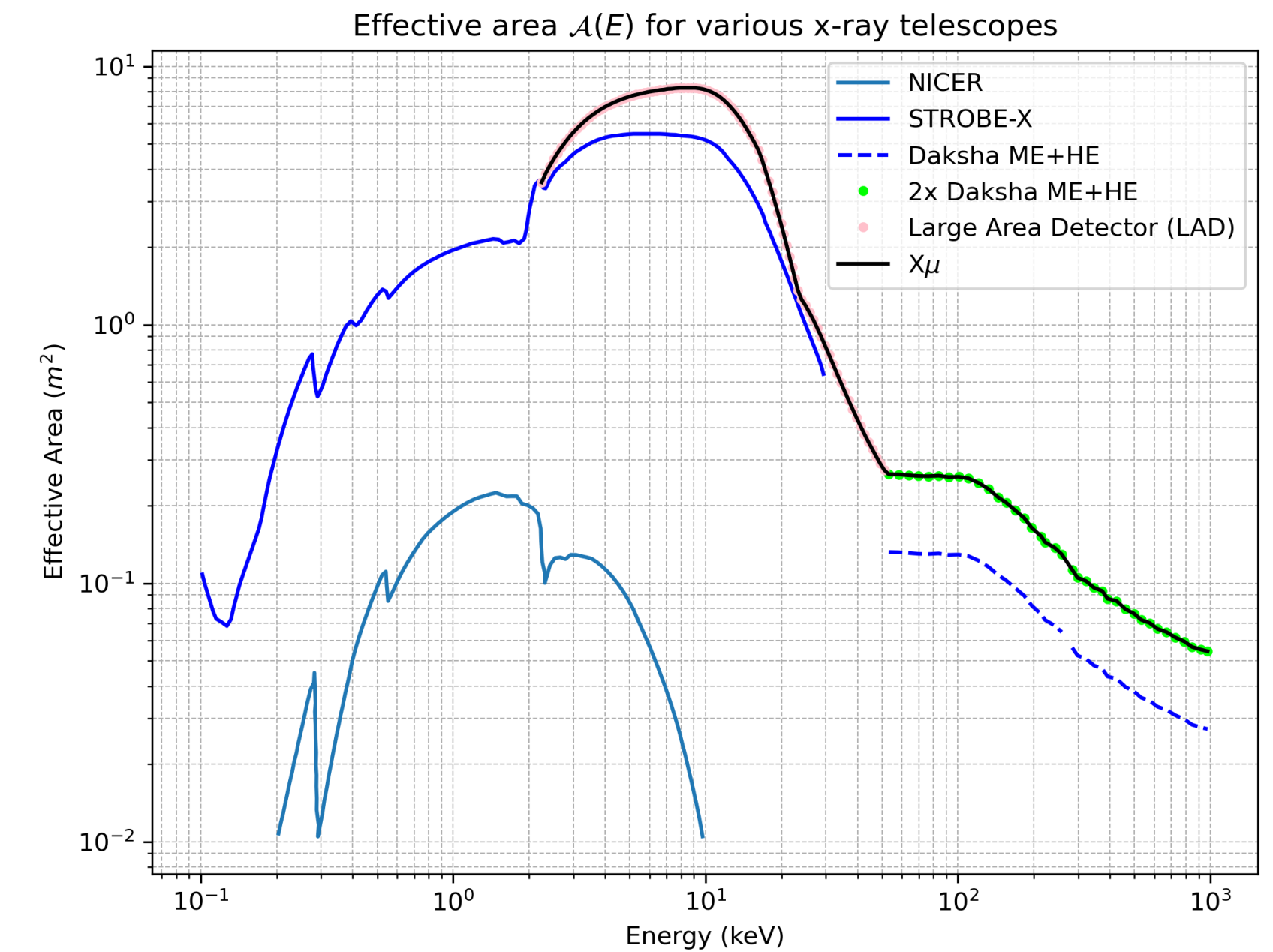
<https://svs.gsfc.nasa.gov/>



**STROBE-X**

arXiv:1903.03035

Expected launch: 2031

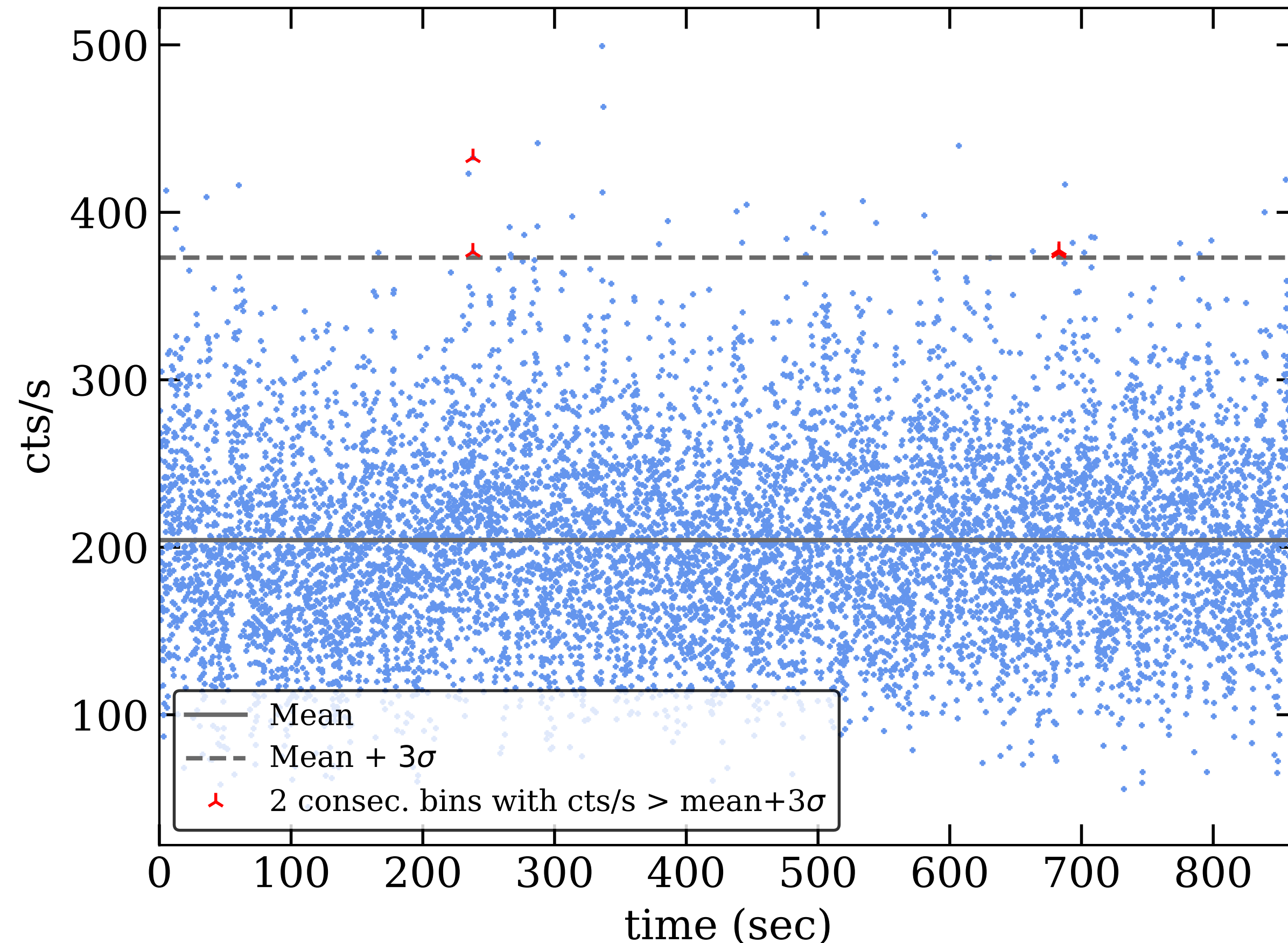


$X\mu$

| x-ray pulsar | net exposure (days) | $D_S$ (kpc) | $(\ell, b)$                    | $\sigma_B/B$ |
|--------------|---------------------|-------------|--------------------------------|--------------|
| SMC X-1      | 1.74                | 64 [28]     | $(300.41^\circ, -43.56^\circ)$ | 0.28         |
| Cyg X-2      | 5.47                | 11 [29]     | $(87.33^\circ, -11.32^\circ)$  | 0.02         |
| Vela X-1     | 4.46                | 2 [30]      | $(263.06^\circ, 3.93^\circ)$   | 0.25         |
| Crab pulsar  | 4.76                | 2 [31]      | $(184.56^\circ, -5.78^\circ)$  | 0.01         |



# Statistical way of interpreting X-ray pulsar light curve



How to identify a true microlensing event in the light curve?

**Excess x-ray photon counts at  $3\sigma$  level for 3 consecutive time bins is a detection!**

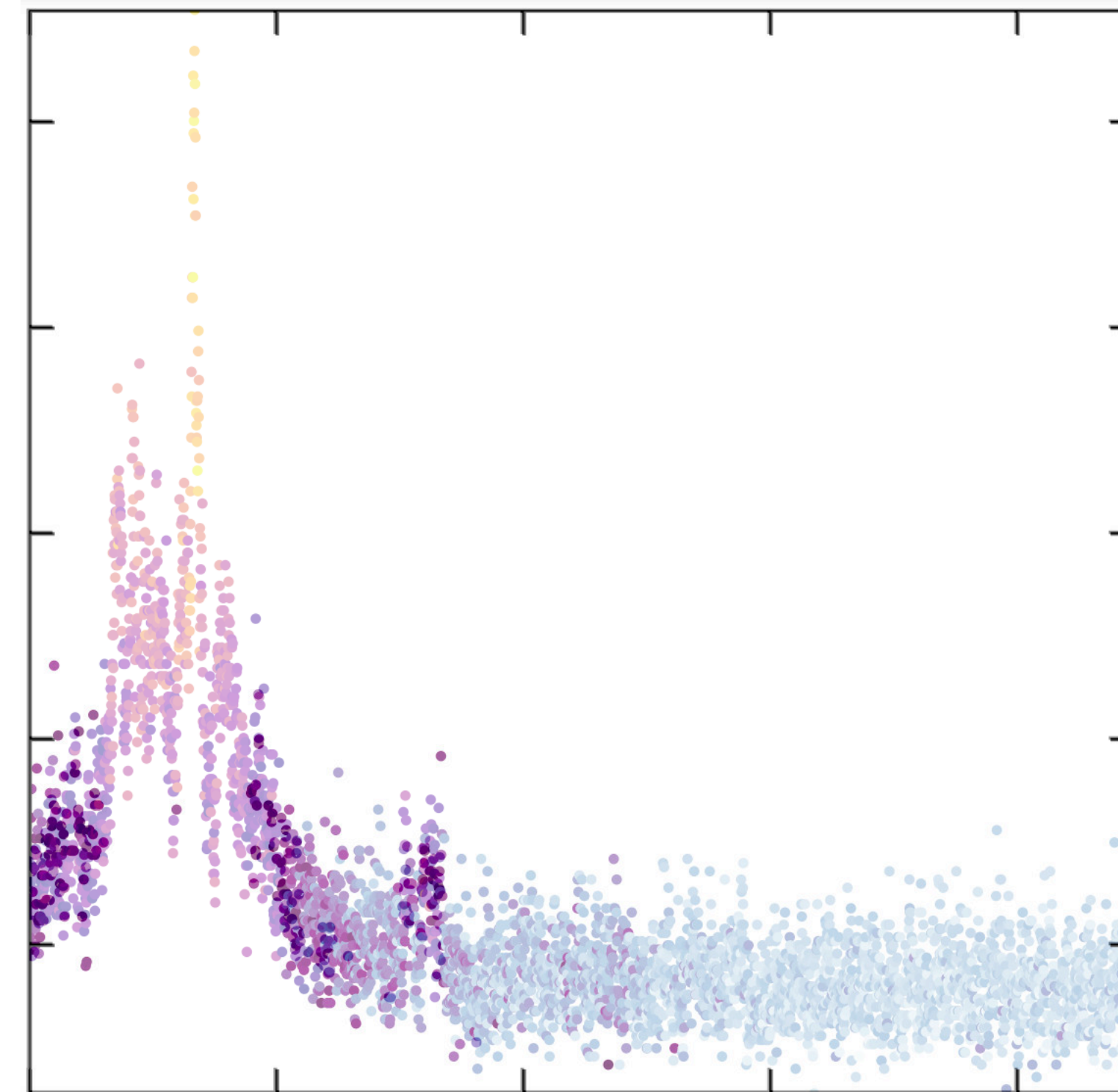
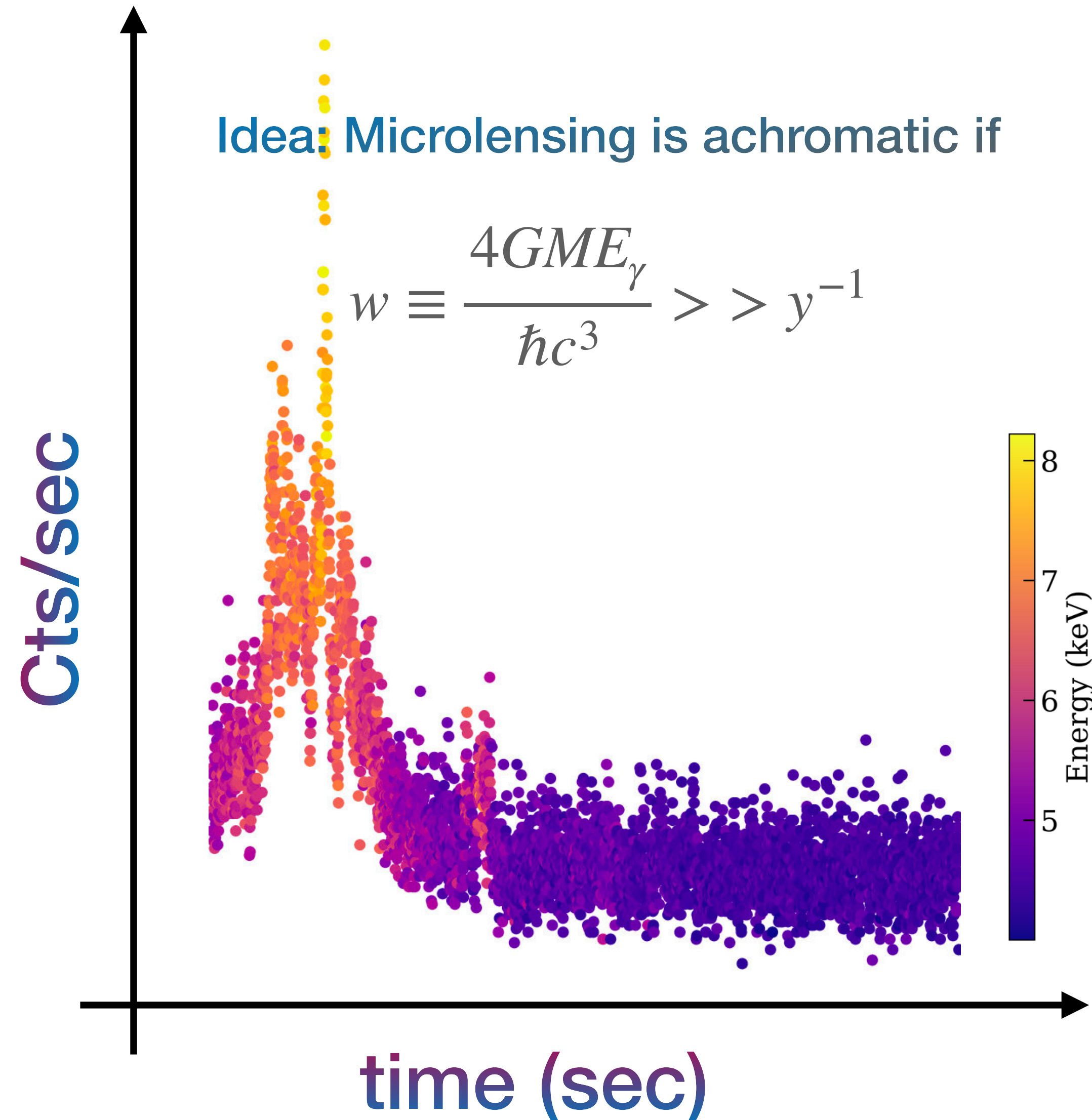
+

**Achromaticity ( $w \gtrsim y^{-1}$ )**

$$\mathbb{P}(N_\sigma) = [1 - \Phi(N_\sigma = 3)]^{N_{consec}=3} = 2.5 \times 10^{-9}$$



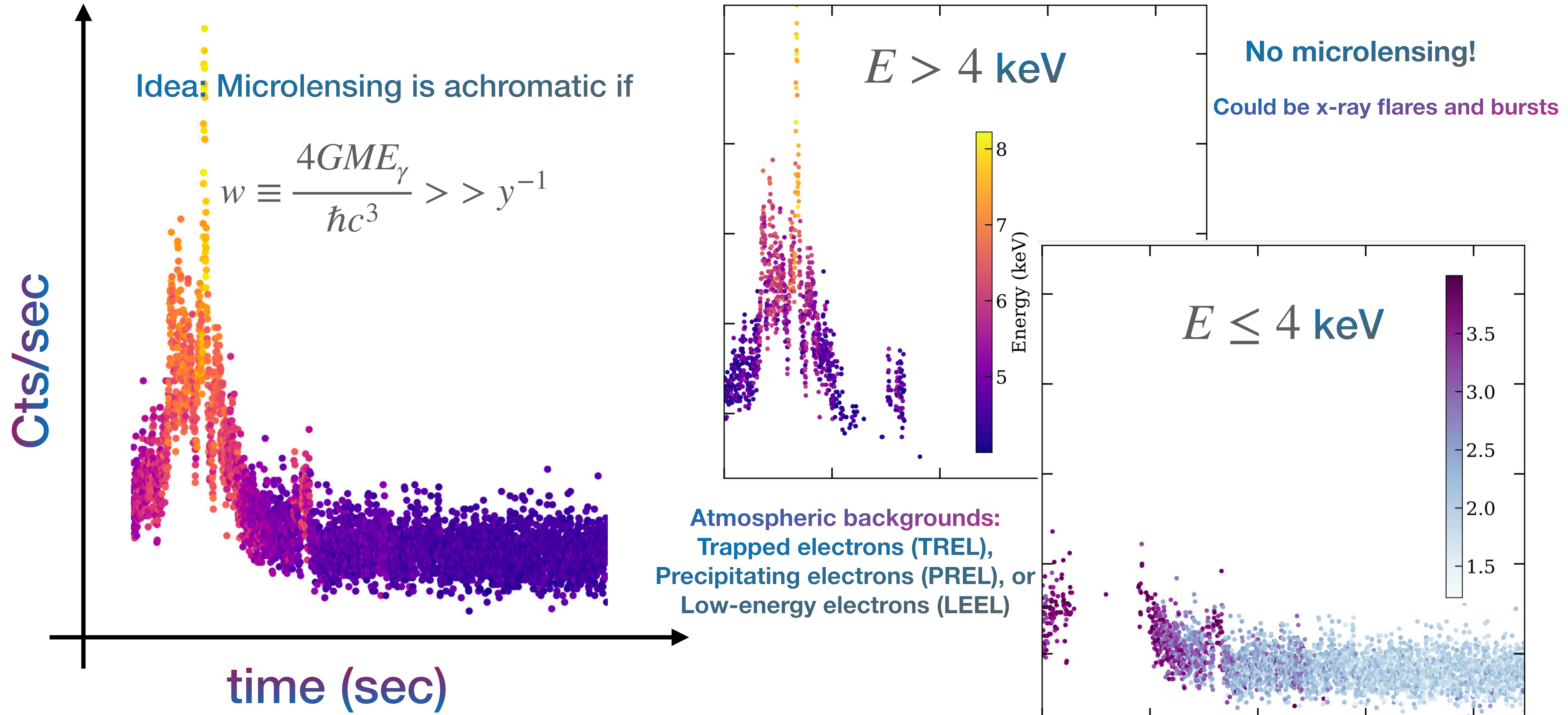
# A new diagnostic for x-ray microlensing



Split the light  
curve into  
multiple energy  
windows



# A new diagnostic for x-ray microlensing





# Microlensing Event Rate

$$\frac{d\Gamma}{dt_E} = f_{PBH} \frac{2D_s}{Mv_0^2} \int_0^1 dx \rho_{dm}(x) v_E^4(x) e^{-v_E^2/v_0^2} \times \int_0^{y_T(x)} \frac{dy}{\sqrt{y_T^2 - y^2}} \times \mathbb{P}(n_\sigma)$$

Local DM  
density

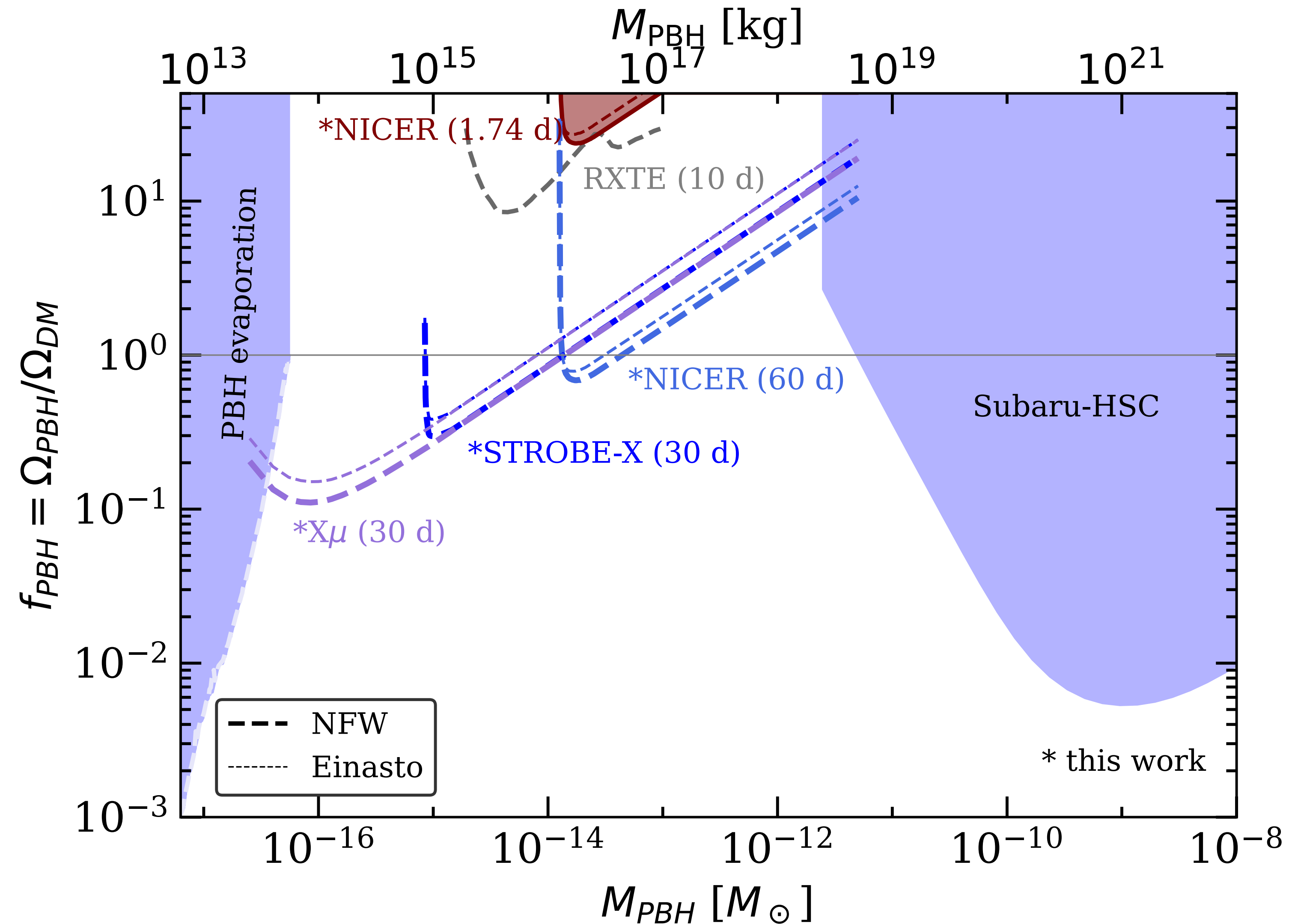
Remember

$$d\Gamma \sim 1/\sqrt{M}$$

$$v_E \equiv 2r_E \sqrt{y_T^2 - y^2} / t_E$$

Total # of microlensing events:  $N_{ev} = f_{PBH} \sum_i T_{obs}^i \int_{t_{min,i}}^{t_{max,i}} dt_E \frac{d\Gamma_i}{dt_E}$

# X-ray telescope sensitivities for PBH dark matter detection/exclusion



$X\mu$ : used the SED of  
Crab-like pulsar in  
SMC



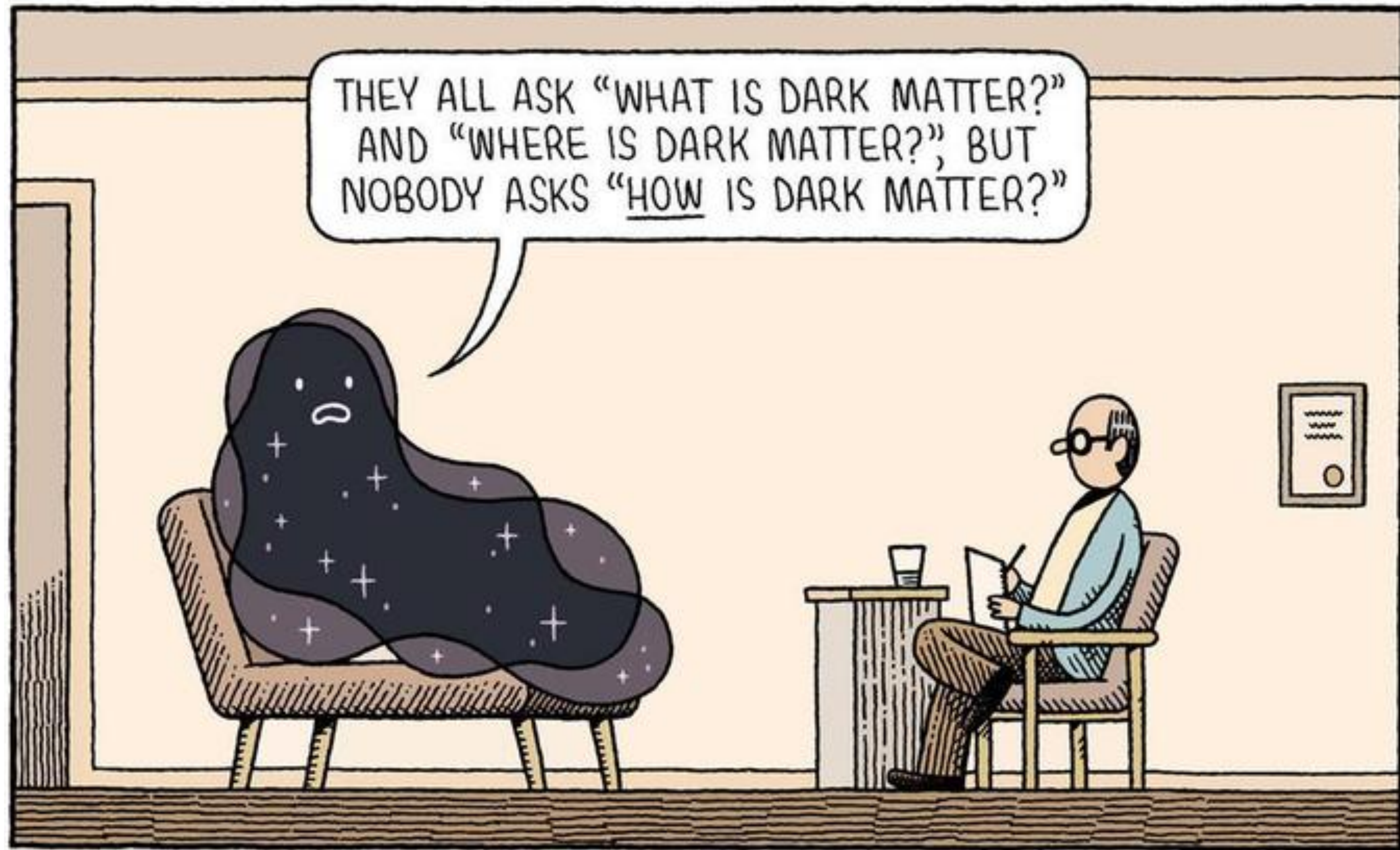
# Take Home

- Primordial black holes (PBHs) in the mass range  $10^{-16} - 10^{-11} M_{\odot}$  may constitute 100% dark matter.
- **WO microlensing of bright x-ray pulsars** provide the most robust and immediately implementable opportunity to uncover PBH dark matter.

# Take Home

- Primordial black holes (PBHs) in the mass range  $10^{-16} - 10^{-11} M_{\odot}$  may constitute 100% dark matter.
- **WO microlensing of bright x-ray pulsars** provide the most robust and immediately implementable opportunity to uncover PBH dark matter.
- **NICER telescope** can probe this window near  $10^{-14} M_{\odot}$  with **just two months of exposure on the x-ray pulsar SMC-X1!**
- **PBH evaporation limit** at around  $10^{-16} M_{\odot}$ , may be probed with a minimal microlensing setup involving **hard x-ray pulsars!**

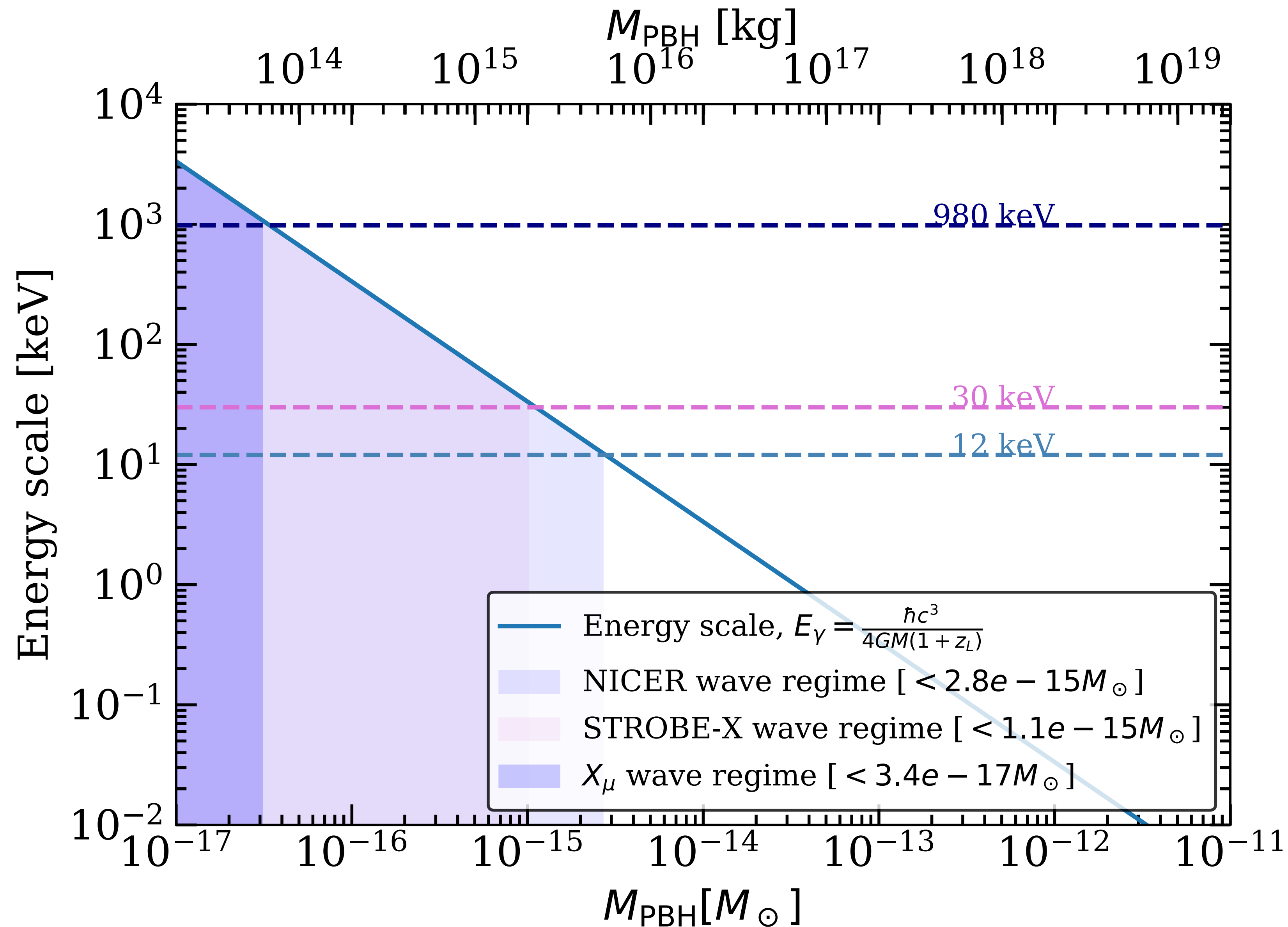




TOM GAULD for NEW SCIENTIST

Thank you for listening :)

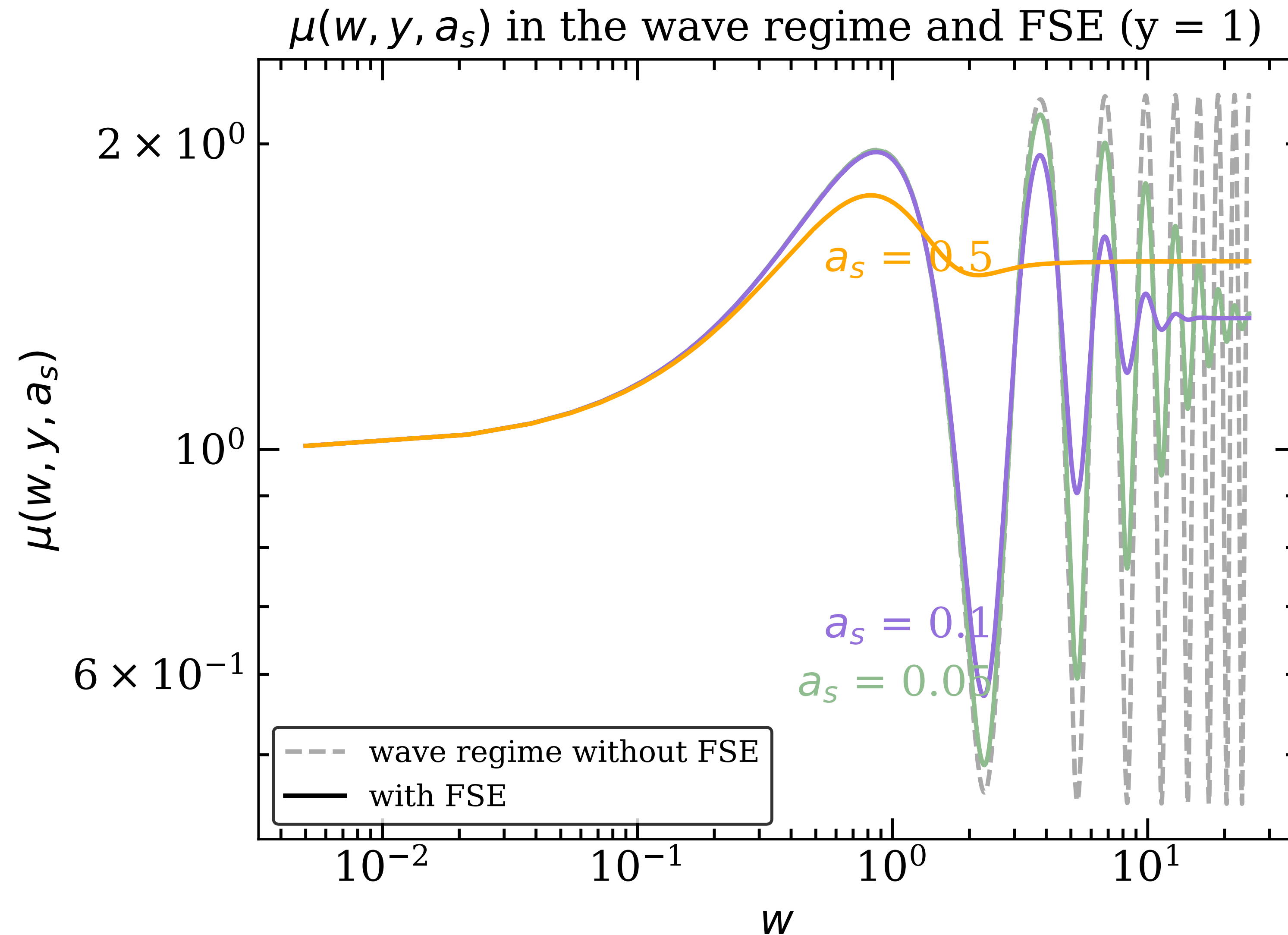




**PBH masses  
that can be  
probed**



# Buffer slides: Frequency dependent magnification



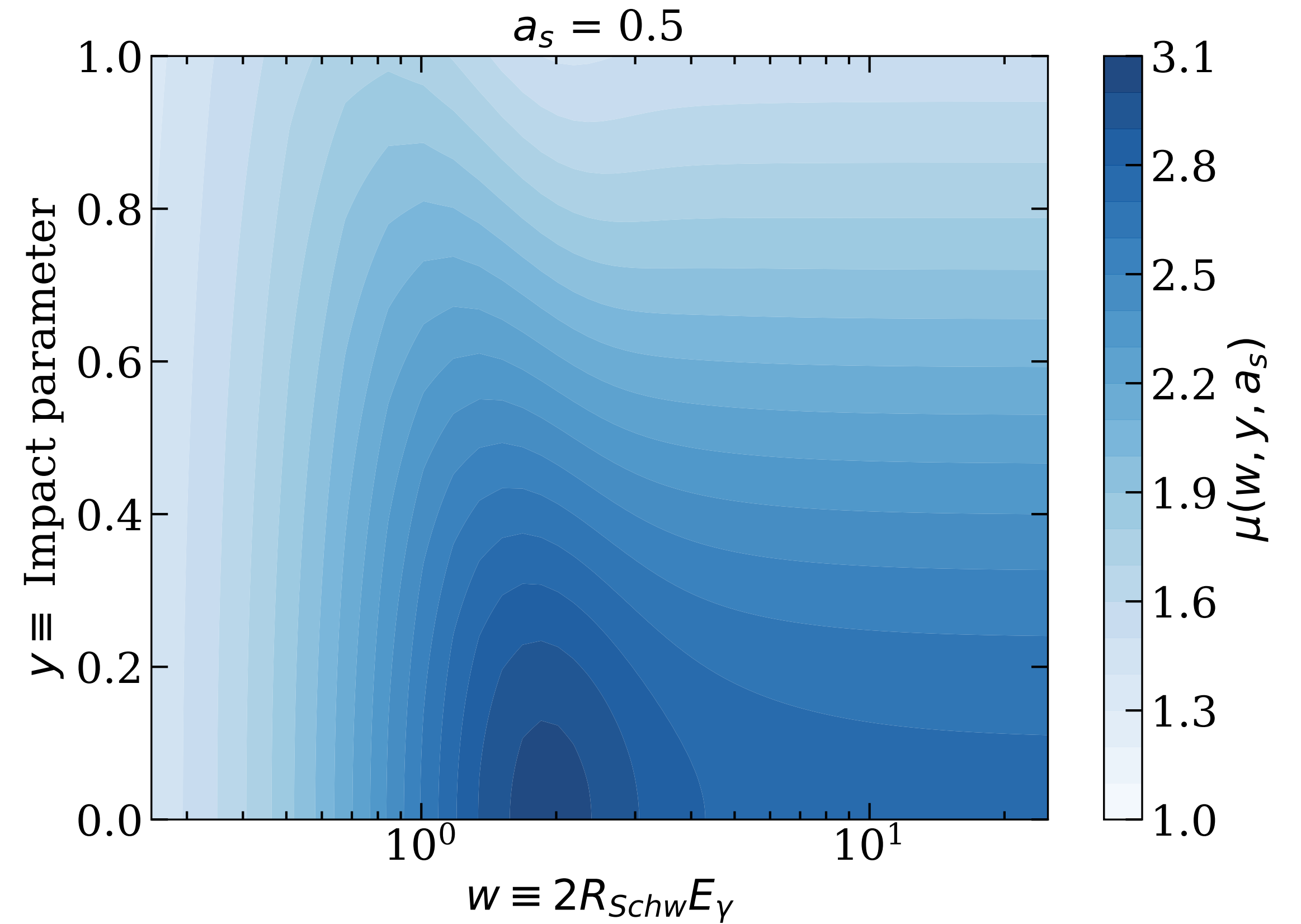
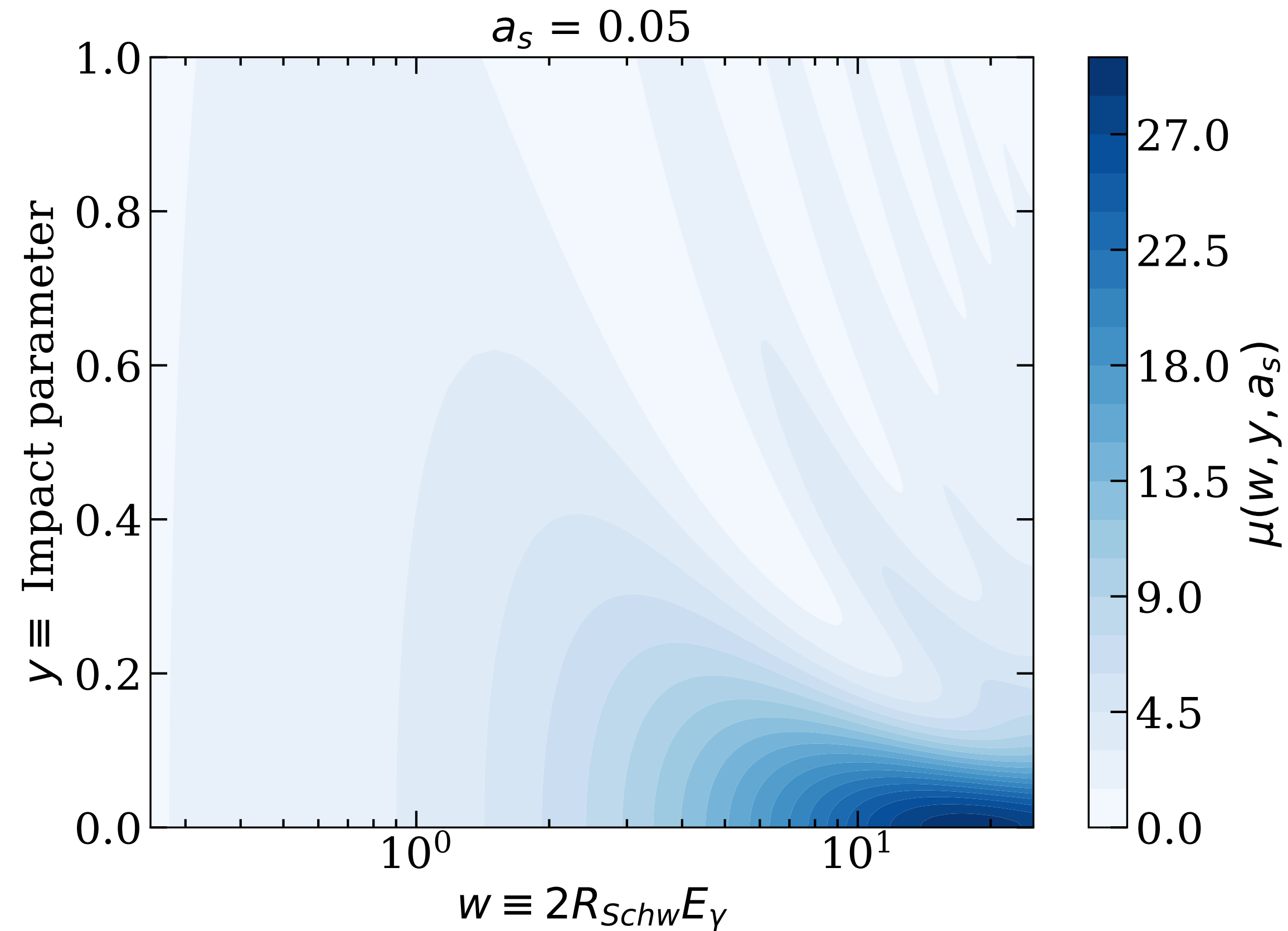
**Microlensing magnification  
in the wave regime**

+

**finite extent of the source**

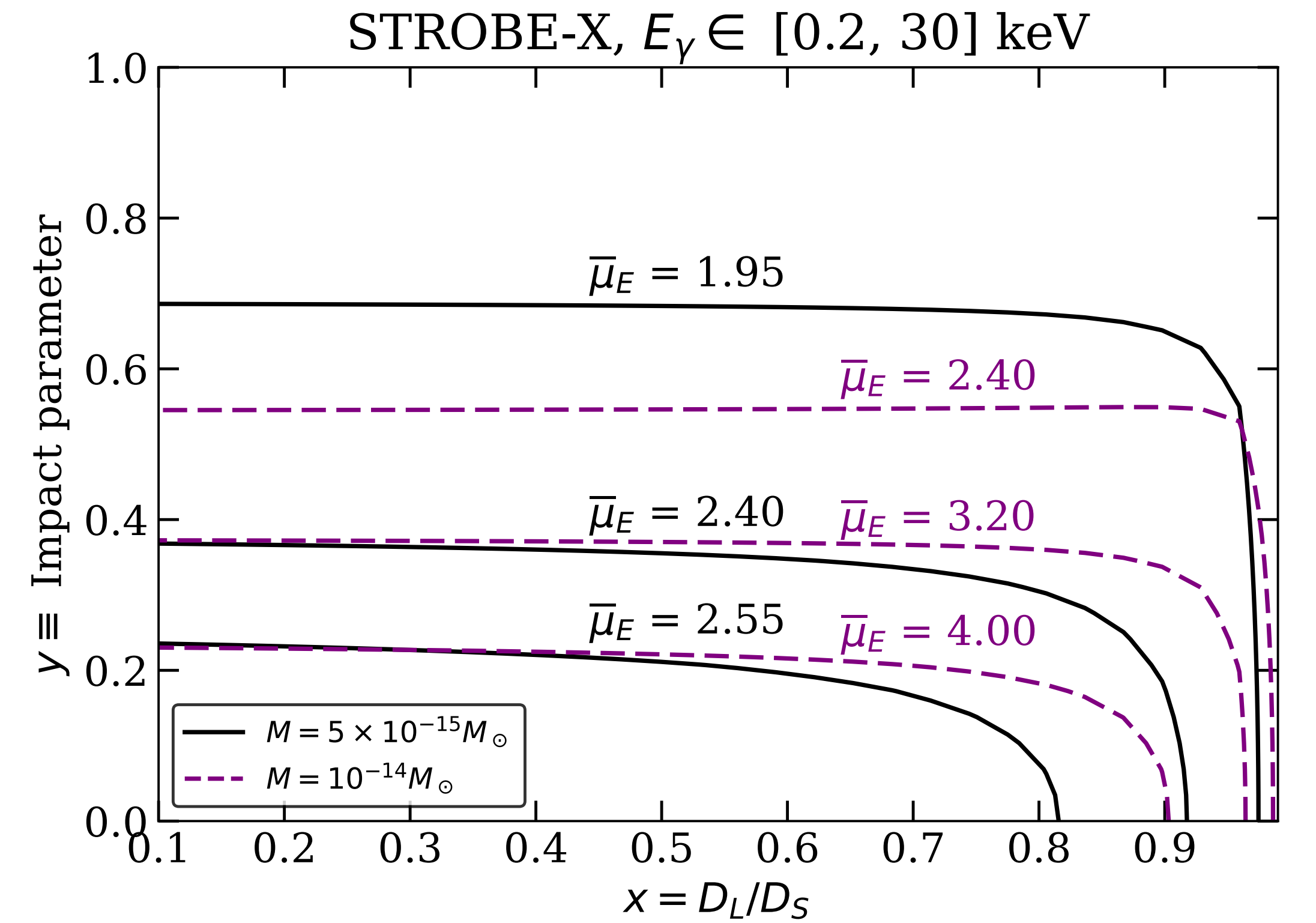
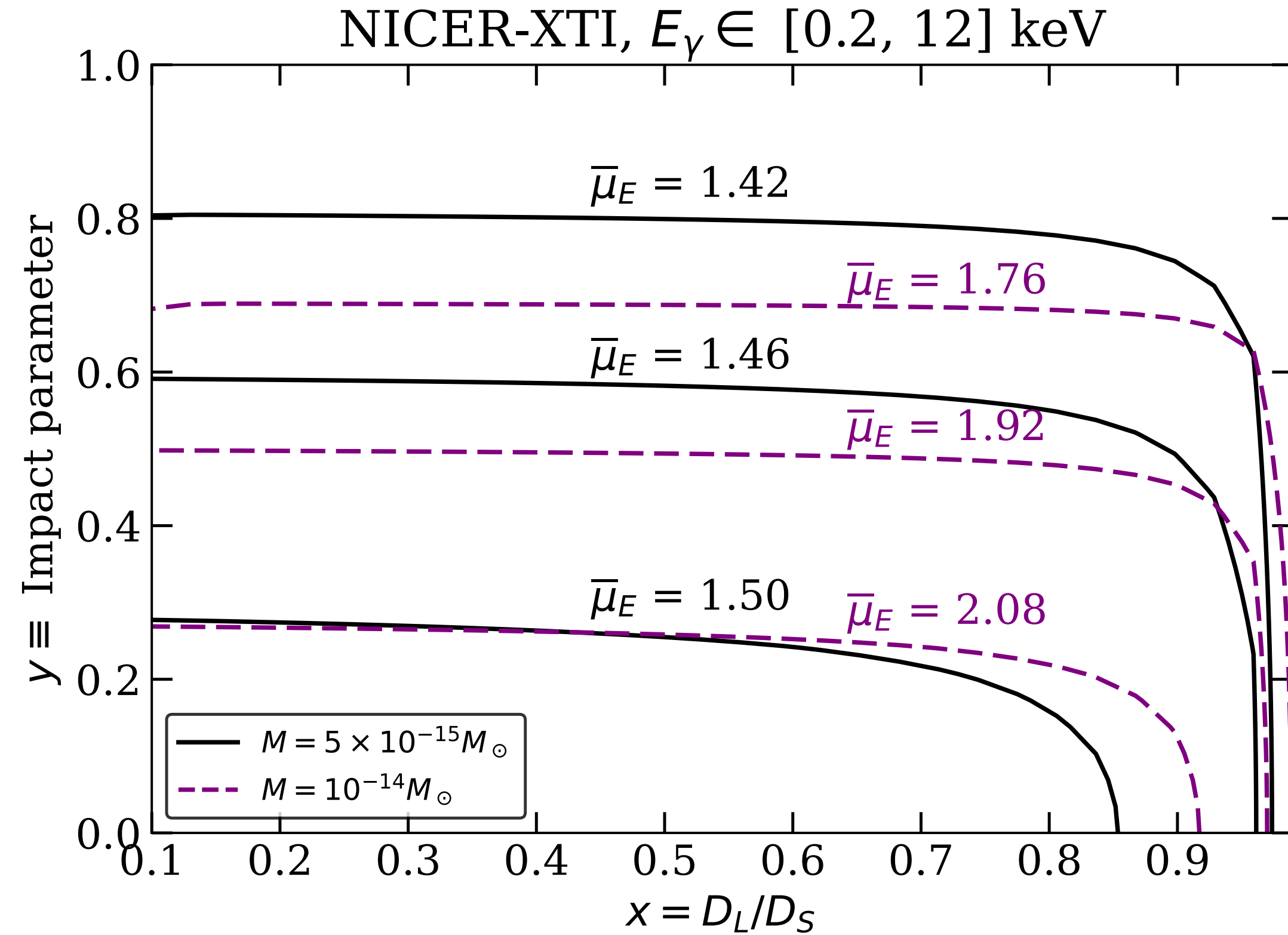
$$a_s = \frac{xR_s}{r_E}$$

# Buffer slides: Magnification factor $\mu(w, y, a_s(x))$ in w-y space





# Buffer slides: Detector and pulsar specific energy averaged magnification



# Buffer slides: Statistical way of interpreting X-ray pulsar light curve

Threshold magnification:

$$\mu_{thresh} = \frac{1 + N_{\sigma}\sigma_B/B}{1 + n_{\sigma}\sigma_B/B}$$

Detection criterion:

$$\bar{\mu}_E \gtrsim \mu_{thresh}$$

$N_{\sigma} - N_{consec}$  criterion:

$N_{\sigma} = 3$  and  $N_{consec} = 3$  with  $n_{\sigma} = -1$  are optimal choices.

- Remark:  $\mathbb{P}(n_{\sigma} = 0) = 0.13$ ; low event acceptance
- $\mathbb{P}(n_{\sigma} = -1) = 0.596$ ; optimal yet significant choice
- $\mathbb{P}(n_{\sigma} = -2) = 0.93$ ; too aggressive choice



# Buffer slides: Condition for time binning of light curve

- Killing the Look elsewhere effect

$$\mathbb{P}(N_\sigma) < \frac{t_{\text{bin}}}{t_{\text{exp}}}$$

$$\mathbb{P}(N_\sigma) = [1 - \Phi(N_\sigma = 3)]^{N_{\text{consec}}=3} = 2.5 \times 10^{-9}$$

$$\frac{t_{\text{bin}}}{t_{\text{exp}}} = 1.9 \times 10^{-8} \left( \frac{t_{\text{bin}}}{0.1 \text{ s}} \right) \left( \frac{60 \text{ days}}{t_{\text{exp}}} \right)$$